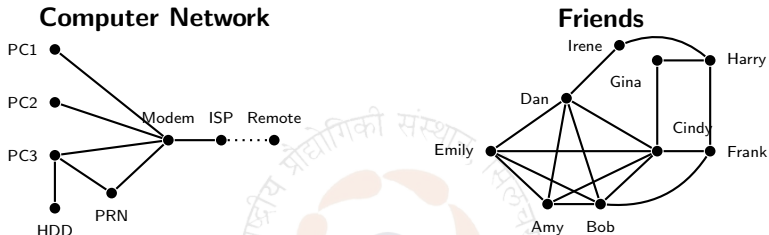


CS-5136: Graph Theory

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Why this course?



We have a network of items and connections between them. Examples:

- Telephone networks, computer networks
- Transportation networks (bus/subway/train/plane)
- Social networks
- Family trees, evolutionary trees
- Molecular graphs (atoms and chemical bonds)
- Various data structures in Computer Science

For whom?

- Anyone who plans and/or schedules.
- Anyone from under-graduate, post-graduate, and doctoral study including practitioners.
- Anyone using combinatorial optimization.
- **Social Media Marketing:** One may target the people who may be interested in your product. How can you define such a target audience?
- **Search Engine Algorithms:** Once a query is made to search a specific set of words and finds millions of matches, how does the engine rank them to show the most popular ones first?
- **Bioinformatics:** Gene regulatory networks and protein-protein interaction networks. What are the functional links between genes or proteins?
- Many more ...

About

- Graphs abstraction & their properties
- Traversal & planarity
- Algebraic representation

Syllabus

Introduction to graphs and their representation, finite and infinite graphs, incidence and degree, path, Directed graph, single source shortest path, all pair shortest path, directed acyclic graph, Euler's graph, Hamiltonian paths and circuits, Basic results of trees, minimum cost spanning tree, Introduction to cut-sets and cut-vertices, connectivity and separability, Basic concepts of vector space of graph, sets with one or two operations, basis vector, circuit and cut-set, subspaces, orthogonal vectors and spaces, Matrix representation of graph, incidence matrix, circuit matrix, path matrix, cut-set matrix and adjacency matrix.

Plan

Grading (As per last semester; might get reviewed by the Competent Authority)

1. Assessment - 20% based on test(s), assignment(s) etc.
2. Mid-term - 30% evaluated based on examination.
3. End-term - 50% evaluated based on examination.

Methodology

- Google Classroom (**Class Code: yedo77yd**) - use Institute e-mail id only.
- Change your name to [Roll No.]-[First Name][Last Name].
- All announcements will be made through G-Classroom only.
- All communication through e-mail only. **Slides will not be shared.**

Academic Honesty & Integrity: All test/project work is entirely your own with no collaboration. Every student will pursue their academic goals honestly. Each student is personally accountable for all his/her submitted work.

Introduction



Problem 1: Railway transportation



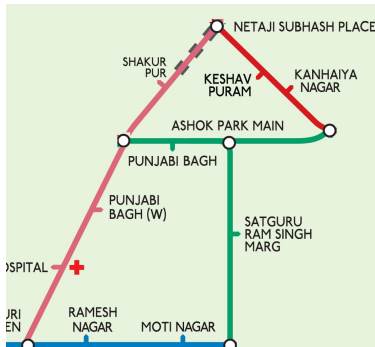
Issue I: New timetable

- Suppose a new timetable was introduced by the Delhi Metro.
- Many new connections and transfer options were introduced, but also with somewhat longer travel and stopping times and discontinuation of some direct connections.
- So, timetable planners need computer algorithms to make better use of the capacity of the railway network.
- Suppose the basis of the timetable is the Basic Hourly Pattern (BUP): the timetable repeats every 60 minutes.
- Each hourly departure and arrival time is represented by a number,
$$\{0, 1, 2, \dots, 59\}$$
- Determining the times in the BUP then comes down to assigning numbers modulo 60 to the “unknowns”.
- What are these unknowns, and what conditions must they satisfy?

Issue II: Minimum circulation of railway stock

- Consider the *Blue Line*. Dwarka Sector 21 - Janakpuri (West) - Rajouri Garden - Mandi House - Noida Electronic City and *vice versa*.
- For each of the stages of any scheduled train, Delhi Metro has determined an *expected number of passengers*, divided into first class and second class.
- Assume the train stock consists of one type of two-way train-units, each consisting of three carriages. Number of seats for each class in a carriage is known.
- Each unit has at both ends a driver's cabin, and units can be coupled together, up to a certain maximum number of units.
- The train length can be changed, by coupling or decoupling units, at the terminal stations of the line.
- For each place X mentioned above, the number of train-units staying overnight at X should be constant during the week.
- What is the minimum number of train-units that should be available to perform the daily cycle of train rides required?

Issue III: Circular railway



- Suppose there is a busy area with lot of commuters.
- Delhi Metro wishes to run circular train on this route.
- The train will stop at each station.
- On each circulation, every segment between a pair of stations should be traversed only once.
- Is it possible? Or, should there be some modification?

Problem 2: DNA Sequencing

- **Sanger Method (1977):**
Labeled ddNTPs terminate DNA copying at random points.
- **Gilbert Method (1977):**
Chemical method to cleave DNA at specific points (G, G+A, T+C, C).



Frederick Sanger

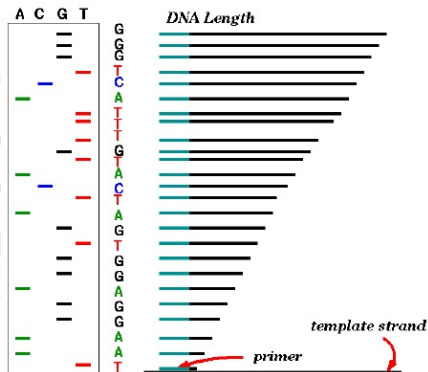
- Both methods generate labeled fragments of varying lengths that are further electrophoresed.



Walter Gilbert

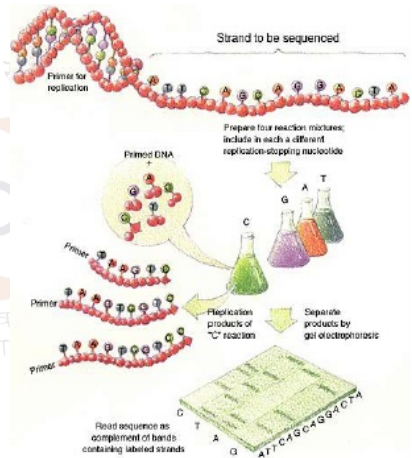
Sanger Method: Generating Read

- Start at primer (restriction site).
- Grow DNA chain.
- Include ddNTPs.
- Stop reaction at all possible points.
- Separate products by length, using gel electrophoresis



Sequencing

- Shear DNA into millions of small fragments.
- Read 500 – 700 nucleotides at a time from the small fragments.



Fragment Assembly

- **Computational Challenge:** assemble individual short fragments (“reads”) into a single genomic sequence (“superstring”).
- Until late 1990s the so called “shotgun fragment assembly” of the human genome was viewed as an intractable problem, because it required so much work for a large genome.
- How to circumvent the computational challenge?

Shortest Superstring Problem (SSP)

- **Problem:** Given a set of strings, find a shortest string that contains all of them.
- **Input:** Strings $s_1, s_2, \dots, s_n$.
- **Output:** A “superstring” s that contains all strings $s_1, s_2, \dots, s_n$ as substrings, such that the length of s is minimized.

SSP: Example

Set of strings: {000, 001, 010, 011, 100, 101, 110, 111}

SSP: Example

Set of strings: {000, 001, 010, 011, 100, 101, 110, 111}

Concatenated superstring: 000 001 010 011 100 101 110 111

SSP: Example

Set of strings: {000, 001, 010, 011, 100, 101, 110, 111}

Concatenated superstring: 000 001 010 011 100 101 110 111

Shortest
superstring

000

001

010

011

100

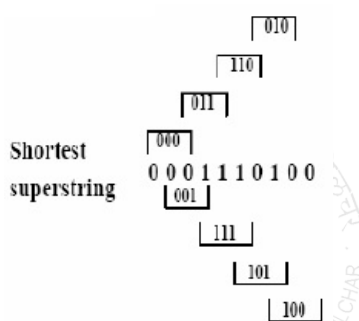
101

110

111

0001110100

SSP: Example



- So the greedy guess of concatenating all the strings together turns out to be substantially suboptimal (length 24 vs. 10).
- Also, this formulation does not take into account the possibility of sequencing errors, and it is difficult to adapt to handle that consideration.
- **Complexity:** NP-Complete.

The Overlap Function

Definition

Given strings s_i and s_j , $overlap(s_i, s_j)$ is defined as the length of the longest prefix of s_j that matches a suffix of s_i .

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Example:

$s_1 = \text{aaaggcatcaaataagggcatcaaa}$

$s_2 = \text{aagcatcaaataagggcatcaaa}$

The Overlap Function

Definition

Given strings s_i and s_j , $overlap(s_i, s_j)$ is defined as the length of the longest prefix of s_j that matches a suffix of s_i .

Example:

$s_1 = \text{aaaggcatcaa} \text{atctaaaggcatcaaa}$

$s_2 = \text{aagcatcaa} \text{atctaaaggcatcaaa}$

aaaggcatcaaatctaaaggcatcaaa

aaaggcatcaaaatctaaaggcatcaaa

Therefore, $overlap(s_1, s_2) = 12$.

SSP, a NP-Complete Problem?

- Construct a graph G as follows:
 - The n vertices represent the n strings $s_1, s_2, \dots, s_n$.
 - For every pair of vertices s_i and s_j , insert an edge of length $overlap(s_i, s_j)$ connecting the vertices.
- Then finding the shortest superstring will correspond to finding the shortest *Hamiltonian path* in G .
- But this is the **Traveling Salesman Problem** (TSP), which is known to be NP-complete.
- **Home work:** Show that any TSP can be formulated as a SSP. (Due: Next week, this class.)

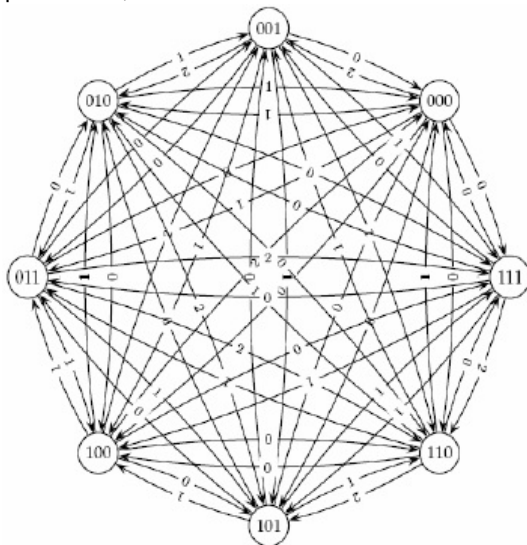
Reducing SSP to TSP: Example 1

Consider the previous set of strings,

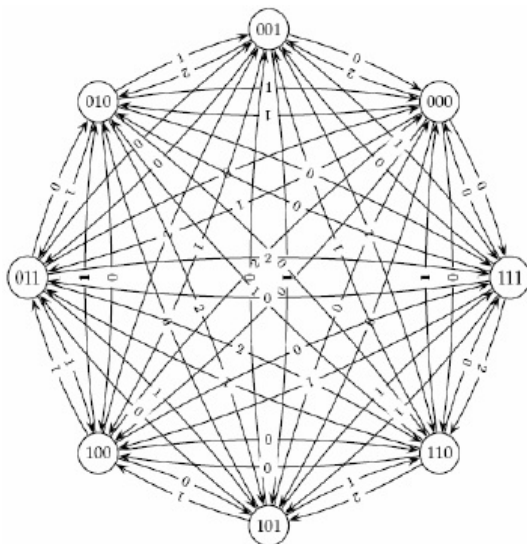
$$S = \{000, 001, 010, 011, 100, 101, 110, 111\}$$

Reducing SSP to TSP: Example 1

Then the graph for S is,



Reducing SSP to TSP: Example 1



One minimal Hamiltonian path gives our previous superstring, 0001110100.

Reducing SSP to TSP: Example 2

Consider the set of strings,

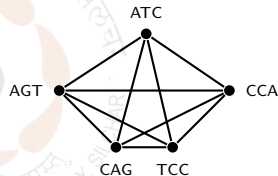
$$S = \{\text{ATC}, \text{CCA}, \text{CAG}, \text{TCC}, \text{AGT}\}$$

Reducing SSP to TSP: Example 2

- Consider the previous set of strings,

$$S = \{ATC, CCA, CAG, TCC, AGT\}$$

- The graph is,

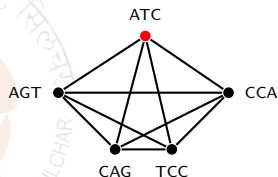


Reducing SSP to TSP: Example 2

- Consider the previous set of strings,

$$S = \{\text{ATC}, \text{CCA}, \text{CAG}, \text{TCC}, \text{AGT}\}$$

- The graph is,
- A minimal Hamiltonian path gives the shortest superstring **ATCCAGT**.



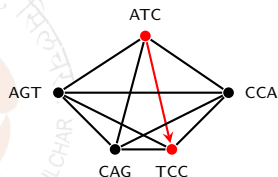
ATC

Reducing SSP to TSP: Example 2

- Consider the previous set of strings,

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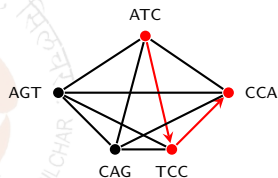
ATCC

Reducing SSP to TSP: Example 2

- Consider the previous set of strings,

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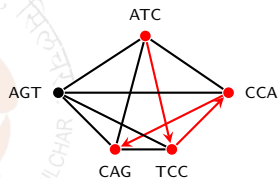
ATCCA

Reducing SSP to TSP: Example 2

- Consider the previous set of strings,

$$S = \{\text{ATC}, \text{CCA}, \text{CAG}, \text{TCC}, \text{AGT}\}$$

- The graph is,
- A minimal Hamiltonian path gives the shortest superstring **ATCCAGT**.



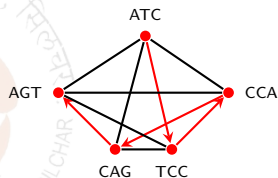
ATCCAG

Reducing SSP to TSP: Example 2

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ATCCAGT

The abstraction



Graph

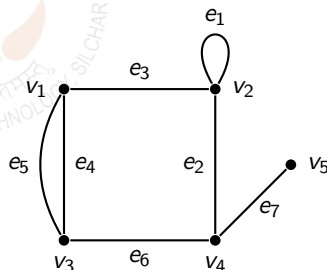
Simply stating, a *graph* is a diagram consisting of vertices and edges. Formally,

Definition (Graph)

A graph $G = (V, E)$ consists of two sets namely, $V = \{v_1, v_2, \dots\}$ called vertices and $E = \{e_1, e_2, \dots\}$ called edges such that every edge is represented as $e_i = (v_p, v_q)$ connecting the vertices v_p and v_q .

An example,

- $V = \{v_1, v_2, v_3, v_4, v_5\}$
- $E = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$
- $|V| = 5, |E| = 7$



G_1

Some basics

- Every edge connects a pair of vertices.
- Traversing a chain of edges, one can reach from one vertex to another non-adjacent vertex.
- An edge can also be represented by a pair of vertices. e_2 is nothing but (v_2, v_4) .
- There can be multiple unique edges between a pair of adjacent vertices, called *parallel edges*. For example, e_4 and e_5 in the graph G_1 .
- There may exist a *self-loop* which enables one to traverse from one vertex to itself without ever transiting any other vertex.

Some definitions

Definition (Adjacent vertices)

Two vertices are said to be *adjacent* to each other if there exists an edge connecting them.

Definition (Incidence)

An *edge* is said to be *incident on a vertex* if the vertex is one of the end point of the edge.

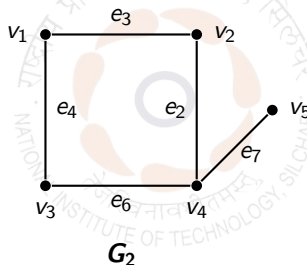
Definition (Degree of a vertex)

Total number of edges incident on a vertex gives its *degree*.

Some more

Definition (Simple graph)

A graph without any parallel edge and self-loops is called *simple graph*.

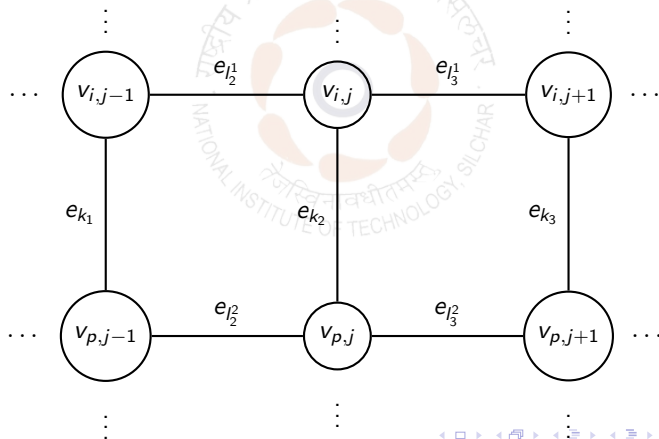


- Graph G_2 is a simple graph having 5 vertices and 5 edges.
- G_2 has *finite* number of vertices and edges.

Some more

Definition (Infinite graph)

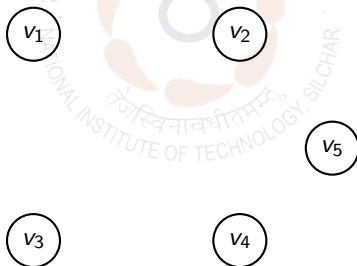
A graph with **infinite number of vertices** is called *infinite graph* otherwise, it is called *finite graph*.



Some more

Definition (Null graph/Trivial graph)

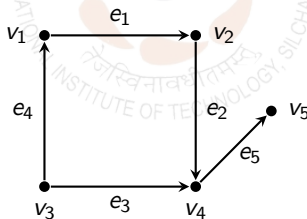
A *null graph* or *trivial graph* is a graph who has non-null vertex set but *null* edge set.



Some more

Definition (Directed graph)

A graph whose every edge is replaced by a directed edge such that traversal from a vertex to another require following direction of the edges, is called *directed graph* or *digraph*.



A few more

Definition (Vertex degree of a directed graph)

Number of edges incident on a vertex of a digraph is called *in-degree* and number of edges emanating from a vertex of digraph is called *out-degree*.

Definition (Isolated vertex)

A vertex of a graph is called *isolated vertex* if there is no edge incident on the vertex.

Definition (Pendent vertex)

A vertex of a graph is called *pendent vertex* if degree of the vertex is 1.

Unless otherwise stated, all discussions are centered around simple undirected graph, in short called, graph.

A few terminologies

Definition

- The cardinality of the vertex-set of a graph G is called the **order** of G and denoted by $|G|$.
- The cardinality of the edge-set of a graph G is called the **size** of G and denoted by $\|G\|$.
- Two vertices incident with the same edge are **adjacent** or **neighbors**.
- Similarly, two edges incident with the same vertex are **adjacent**.
- The number of edges incident with a vertex v of G , with loops counted twice, is the **degree** of v and is written as $d(v)$ or $d_G(v)$.
- The set of neighbors of a vertex v of G , other than v itself, is denoted by $N(v)$ or by $N_G(v)$.

Some results

Theorem (Handshaking Lemma)

$$\sum_{v \in V(G)} d(v) = 2\|G\|$$

Sketchy Proof

1. Each edge contributes +1 degree to each end vertices.
2. Same is true for self-loops also.

Corollary

The number of vertices of odd degree in a graph is always even.

Some results

Theorem

Maximum number of edges in a simple graph of n vertices is $\frac{n(n-1)}{2}$.

Sketchy Proof

1. Proceed by *mathematical induction* on number of vertices.
2. **Basis:** One edge and two vertices.
3. **Hypothesis:** Assume that the statement is true for $n = k$, $k < n$.
4. Without violating the *simple* property of the graph, **argue**.

Home work

Theorem

An infinite graph with **finite number of vertices** will have at least one pair of vertices joined by infinitely many parallel edges.

Due: Next week, this class.

Some more application areas

- **Topological Graph Theory:** Asks questions about methods of embedding graphs into topological spaces (like $\mathbb{R}^2$ or on the surface of a torus) so that certain properties are maintained. For example, the question of planarity asks: Can a graph be drawn on the plane in such a way so that no two edge cross?
- **Graph Coloring:** A question related both to optimization and to planarity asks how many colors does it take to color each vertex (or edge) of a graph so that no two adjacent vertices have the same color. Attempting to obtain a coloring of a graph has several applications to scheduling and computer science.
- **Analytic Graph Theory:** It is the study of randomness and probability applied to graphs. Here, a graph is assumed to be drawn from a probability distribution for studying the properties that certain distributions of graphs may have.
- **Algebraic Graph Theory:** Application of abstract algebra (sometimes associated with matrix groups) to graph theory.

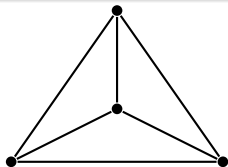
Some more types of graphs

Definition (Complete graph)

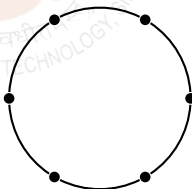
A graph in which every vertex is adjacent to every other vertex, is called a *complete graph*. A complete graph of n vertices is denoted by K_n .

Definition (Cycle graph)

A graph in which every vertex is of degree 2, is called a *cycle graph*. A cycle graph of n vertices is denoted by C_n .



Complete Graph, K_4



Cycle Graph, C_6

Some more types of graphs

Definition (Path graph)

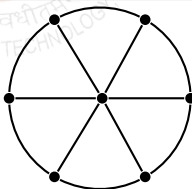
A graph which is obtained by removing one edge from a cycle graph is called *path graph*. A path graph of n vertices is denoted by P_n .

Definition (Wheel graph)

A graph which is obtained by introducing a new vertex in a cycle graph and connecting this vertex to every other vertices of the cycle graph is called *wheel graph*. A wheel graph of n vertices is denoted by W_n .



Path Graph, P_n



Wheel Graph, W_6

Some more types of graphs

Definition (r -regular graph)

A graph in which every vertex has the same degree, r , is called r -regular graph or regular graph.

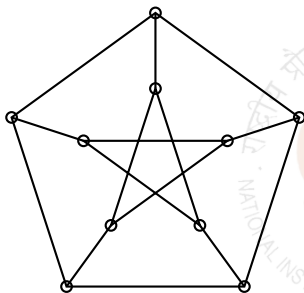
Example

The graph K_4 is an example of 3-regular graph. The graph C_6 is an example of 2-regular graph. Similarly, the **Peterson graph** is also a 3-regular graph.

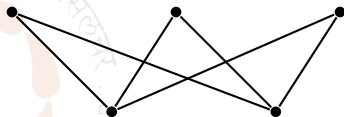
Definition (Bipartite graph)

A graph whose vertex set can be partitioned into two disjoint subsets such that every edge of the graph has one end-vertex in one subset and the other end-vertex in the other subset, is called a *bipartite graph*.

Some more types of graphs



Peterson Graph



Bipartite Graph

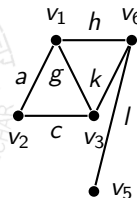
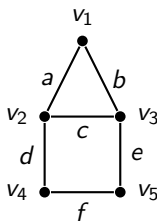
Operations on graphs

- Almost any type of mathematical object has corresponding mathematical operations to manipulate the objects and produce new objects.
- Numbers can be negated, added, or divided.
- Functions can be multiplied or composed. Sets can be complemented or intersected.
- It is natural that analogous operations exist for graphs.
- Graphs can have *unary*, *binary*, and, in some special cases, *m*-ary operations.

Union

Definition (Union)

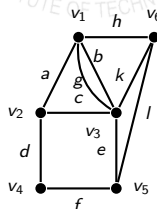
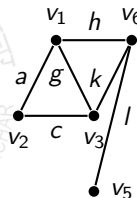
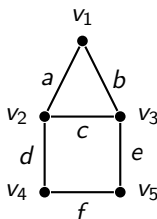
The *union* of two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is another graph G_3 (written as $G_3 = G_1 \cup G_2$) such that, $V_3 = V_1 \cup V_2$ and $E_3 = E_1 \cup E_2$.



Union

Definition (Union)

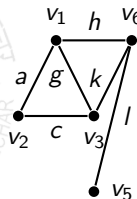
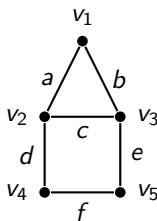
The *union* of two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is another graph G_3 (written as $G_3 = G_1 \cup G_2$) such that, $V_3 = V_1 \cup V_2$ and $E_3 = E_1 \cup E_2$.



Intersection

Definition (Intersection)

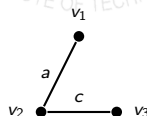
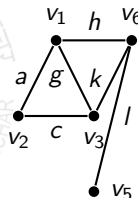
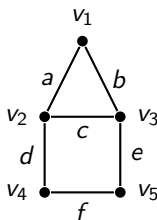
The *intersection*, $G_1 \cap G_2$, of two graphs is another graph G_4 such that, G_4 consists of only of those vertices and edges that are in both G_1 and G_2 .



Intersection

Definition (Intersection)

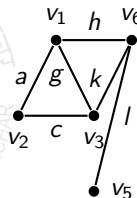
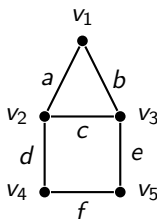
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Ring sum

Definition (Ring sum)

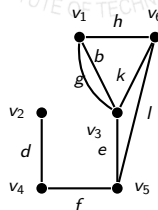
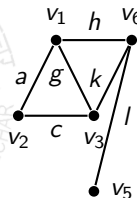
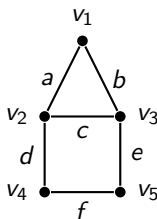
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Ring sum

Definition (Ring sum)

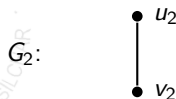
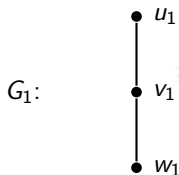
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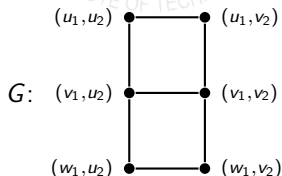
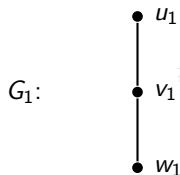
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Algebraic treatment of graph require introduction of two key concepts.

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Two graphs G_1 and G_2 are said to be *isomorphic* if there exists a one-to-one correspondence between their set of vertices and set of edges, respectively, such that incidence relationship is preserved.



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An isomorphism from graph G to graph H is a bijection $\phi : V(G) \rightarrow V(H)$ such that $uv \in E(G)$ if and only if $\phi(u)\phi(v) \in E(H)$. We say G is *isomorphic* to H and write $G \cong H$ for labeled graphs or $G = H$ for unlabeled graphs. Two graphs are *nonisomorphic* if they are not isomorphic.

Isomorphism

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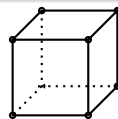
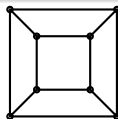
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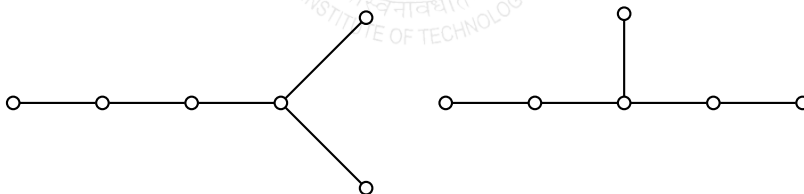


Isomorphism

The definition of isomorphism immediately tells that two isomorphic graphs must have,

- Same number of vertices.
- Same number of edges.
- Equal number of vertices with same degree.

Above conditions are necessary for graph isomorphism but not sufficient .



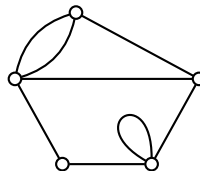
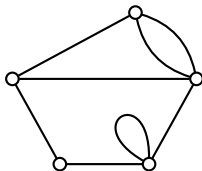
Isomorphism

Example

Both the graphs in the last slide have 6 vertices and 5 edges. However, looking at the incidence relationship of vertex of degree 3 in both the graphs, edges are incident upon this vertex from vertices of degree $\{2, 1, 1\}$ in the LHS graph whereas, edges are incident from vertices of degree $\{2, 2, 1\}$ in the RHS graph. Thus incidence relationship is not preserved and hence, they are not isomorphic.

Testing of graph isomorphism, in general, is a *NP-complete* problem.

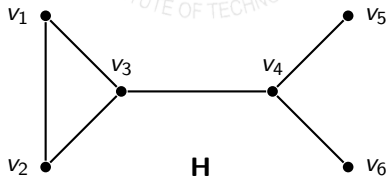
Homework: What can be stated about the graphs below? Due: Next week, this class.



Automorphism

- It may be quite difficult to determine that two isomorphic graphs G_1 and G_2 are in fact isomorphic if G_1 and G_2 are drawn differently or labeled differently.
- However, there is no difficulty in showing that G_1 and G_2 are isomorphic if they are drawn and labeled identically.
- In such case, given a single graph G , the identity function $\phi : V(G) \rightarrow V(G)$ where, $\phi(v) = v$ for all $v \in V(G)$, is an isomorphism.

Consider the graph H below,



Automorphism

- The mapping $\phi_1 : V(H) \rightarrow V(H)$ where, $\phi_1(v) = v$ for all $v \in V(H)$, is an isomorphism.

- Another isomorphism is,

$$\phi_2(v_1) = v_2, \phi_2(v_2) = v_1 \text{ and } \phi_2(v) = v \text{ if } v \neq v_1, v_2$$

- Yet, another isomorphism is,

$$\phi_3(v_5) = v_6, \phi_3(v_6) = v_5 \text{ and } \phi_3(v) = v \text{ if } v \neq v_5, v_6$$

- Finally, last possible isomorphism is,

$$\begin{aligned} \phi_4(v_1) &= v_2, \phi_4(v_2) = v_1, \phi_4(v_5) = v_6, \\ \phi_4(v_6) &= v_5 \text{ and } \phi_4(v) = v \text{ if } v = v_3, v_4 \end{aligned}$$

Automorphism

Definition (Automorphism)

An isomorphism from a graph G to itself is called an *automorphism* of G .

- Since composition is associative, the identity function is an automorphism.
- The inverse of an automorphism is an automorphism.
- Composition of two automorphisms is an automorphism

Definition (Automorphism group)

The set of all automorphisms of a graph G forms a group under the *operation of composition*. This group is denoted by $\text{Aut}(G)$ and is called the **automorphism group** of G .

Example

For the graph H , $\text{Aut}(H) = \{\phi_1, \phi_2, \phi_3, \phi_4\}$.

Automorphism

- Every automorphism of a graph is a *permutation*¹ on $V(G)$.
- Therefore, automorphisms can be expressed in terms of permutation cycles.
- The permutation and the group table for the automorphism group of the graph H is,

$$\phi_1 = \epsilon$$

$$\phi_2 = (v_1, v_2)$$

$$\phi_3 = (v_5, v_6)$$

$$\phi_4 = (v_1, v_2)(v_5, v_6)$$

	ϕ_1	ϕ_2	ϕ_3	ϕ_4
ϕ_1	ϕ_1	ϕ_2	ϕ_3	ϕ_4
ϕ_2	ϕ_2	ϕ_1	ϕ_4	ϕ_3
ϕ_3	ϕ_3	ϕ_4	ϕ_1	ϕ_2
ϕ_4	ϕ_4	ϕ_3	ϕ_2	ϕ_1

¹For a nonempty set A , a bijective function $f : A \rightarrow A$ is a **permutation** of A .

Orbit

Definition (Orbit)

The set of all vertices into which a vertex $v \in V(G)$ can be mapped by some automorphism of G is an *orbit* of G . Two vertices u and v are *similar* if they belong to the same orbit.

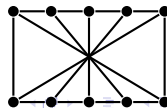
- If a graph G of order n has n distinct orbits, then $\text{Aut}(G)$ consists of a single automorphism, namely, the identity automorphism.
- On the other hand, if a graph G contains a single orbit, then every two vertices of G are similar and G is called **vertex-transitive**.

Theorem

Every vertex-transitive graph is regular but the converse is not true.

Homework: Due next week, this class.

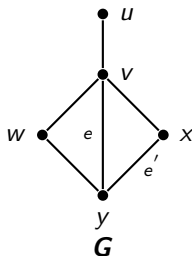
A vertex-transitive graph



Subgraph

Definition (Subgraph)

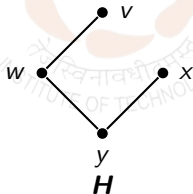
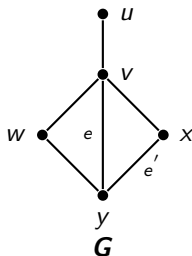
Let $G = (V, E)$ be a graph. A graph $H = (V', E')$ is a *subgraph* of G if $V' \subseteq V$ and $E' \subseteq E$. The subgraph H is *proper* if $V' \subset V$ or $E' \subset E$.



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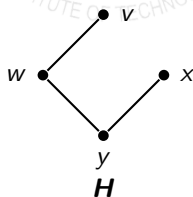
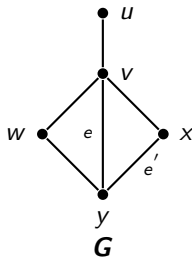
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Definition (Spanning subgraph)

Let $G = (V, E)$ be a graph and $H = (V', E')$ is a subgraph of G . If $V' = V$ then subgraph H is called *spanning subgraph* of G .



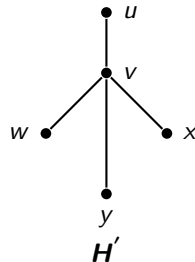
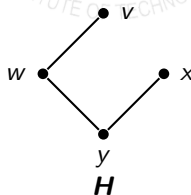
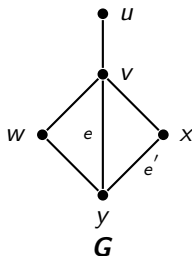
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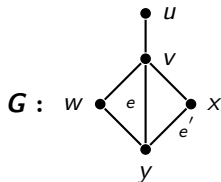
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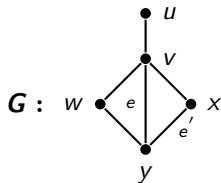
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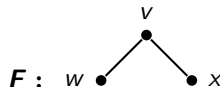
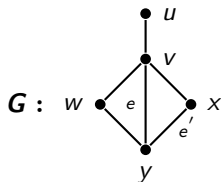
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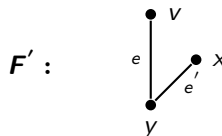
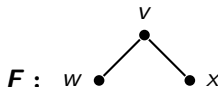
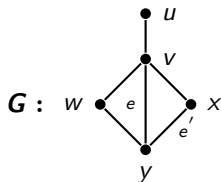
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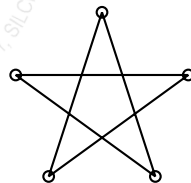
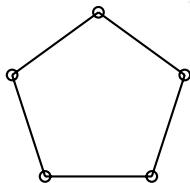
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More operations

Definition (Complement)

Let $G = (V, E)$ be a graph. *Complement* of G , denoted by $\overline{G}$, is obtained by joining each pair of vertices (u, v) by an edge such that, $uv \notin E(G)$ and removing each edge $uv \in E(G)$, but not removing the vertices, for all $u, v \in V(G)$.



A graph isomorphic to its complement.

More operations

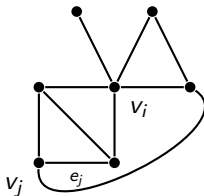
Definition (Decomposition)

A graph G is said to have been decomposed into two subgraphs g_1 , and g_2 if,

$$g_1 \cup g_2 = G$$

$$g_1 \cap g_2 = \text{a null graph}$$

Every edge of G occurs either in g_1 or in g_2 , but not in both.



More operations

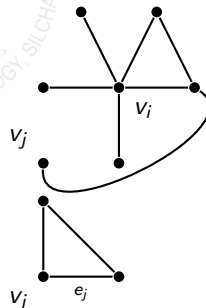
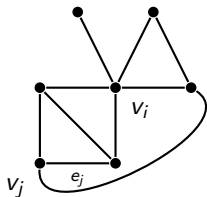
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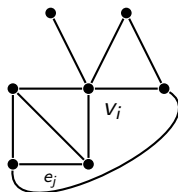
More operations

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If v_i is a vertex in graph G , then $G - v$ denotes a subgraph of G obtained by *deleting* (i.e. removing) v_i from G .

If e_j is an edge in G , then $G - e_j$ is a subgraph of G obtained by *deleting* e_j from G .

Deletion of a vertex is deletion of all edges incident on that vertex. But, $G - e_j = G \oplus e_j$.



G

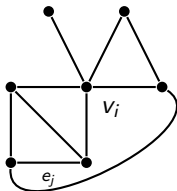
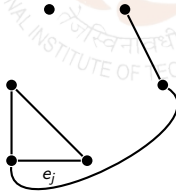
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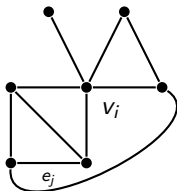
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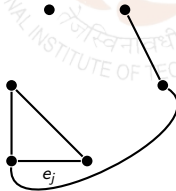
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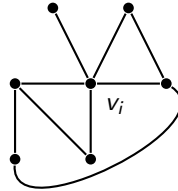
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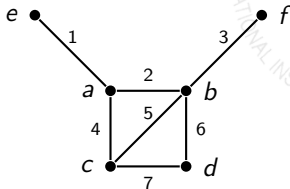


$G - e_j$

More operations

Definition (Fusion)

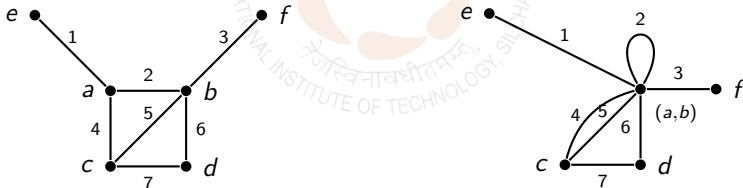
A pair of vertices a, b in a graph are said to be *fused* (*merged* or *identified*) if the two vertices are replaced by a single new vertex such that every edge that was incident on either a or b or on both, is now incident on the new vertex.



More operations

Definition (Fusion)

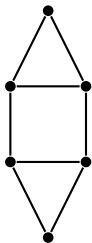
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More operations

Definition (Closure)

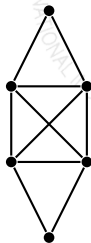
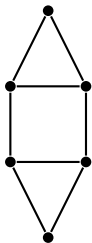
Closure, also sometimes called *Hamiltonian closure*, $C(G)$ of a graph G of order n is the graph obtained from G by recursively joining pairs of nonadjacent vertices whose degree sum is at least n until no such pair remains.



More operations

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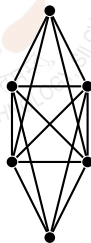
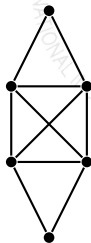
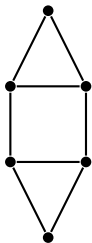
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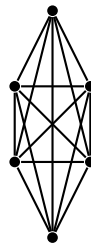
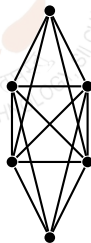
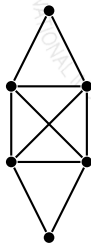
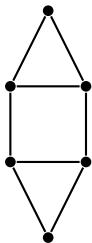
Closure, also sometimes called *Hamiltonian closure*, $C(G)$ of a graph G of order n is the graph obtained from G by recursively joining pairs of nonadjacent vertices whose degree sum is at least n until no such pair remains.



More operations

Definition (Closure)

Closure, also sometimes called *Hamiltonian closure*, $C(G)$ of a graph G of order n is the graph obtained from G by recursively joining pairs of nonadjacent vertices whose degree sum is at least n until no such pair remains.



Home work

Lemma

A graph containing m edges, $\{e_1, e_2, \dots, e_m\}$, can be decomposed into $2^{m-1} - 1$ different ways into pairs of subgraphs g_1 and g_2 .

Due next week, this class.

Vertex Degrees

Degrees

Definition (Minimum degree)

The *minimum degree* of a graph is the minimum degree among the vertices of the graph, G , and is denoted by $\delta(G)$.

Definition (Maximum degree)

The *maximum degree* of a graph is the maximum degree among the vertices of the graph, G , and is denoted by $\Delta(G)$.

For a graph G of order n and v is any vertex of G ,

$$0 \leq \delta(G) \leq \deg(v) \leq \Delta(G) \leq n - 1$$

Definition (Connected graph)

A graph without any isolated vertex is called *connected graph*.

Connected graph

Theorem

Let G be a simple graph with at least two vertices. Then, there exist two distinct vertices u and v of G that have the same degree.

Sketchy Proof

1. Suppose, degrees of all vertices of G , of order n , are distinct.
2. Consider the injective map,

$$\begin{aligned}\deg: V(G) &\rightarrow \{0, 1, 2, \dots, n-1\} \\ v &\mapsto \deg(v)\end{aligned}$$

3. By the pigeonhole principle, it takes both 0 and $n-1$ as values.
4. Say, u is of degree 0 and v is of $n-1$. Are these two vertices adjacent?

Average degree

Theorem

Every graph with average degree at least $2k$, where k is a positive integer, has an induced subgraph with minimum degree at least $k + 1$.



Average degree

Theorem

Every graph with average degree at least $2k$, where k is a positive integer, has an induced subgraph with minimum degree at least $k + 1$.

Sketchy Proof

1. Suppose, average degree G , $d(G) \geq 2k$, and F is an induced subgraph of G with the largest possible average degree and the smallest number of vertices.
2. This is clearly true if $|V(F)| = 1$. So, assume $|V(F)| > 1$
3. Say, $d_F(v) \leq k$, $v \in V(F)$. Obtain a graph $F' = F - v$ which also an induced subgraph of G . Clearly,

$$d(F') = \frac{2|E(F')|}{|V(F')|} \geq \frac{2(|E(F)| - k)}{|V(F)| - 1} \geq \frac{2|E(F)| - d(G)}{|V(F)| - 1} \geq \frac{2|E(F)| - d(F)}{|V(F)| - 1} = d(F)$$

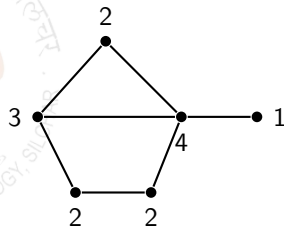
4. Because, $|V(F')| < |V(F)|$, a contradiction.

Degree sequence

Definition (Degree sequence)

When degrees of the vertices of a graph G are listed in a sequence s , then s is called a *degree sequence* of G .

Degree sequence: 4, 3, 2, 2, 2, 1



Definition (Graphical)

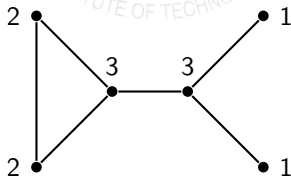
A finite sequence of nonnegative integers is called *graphical* if it is a degree sequence of some graph.

Examples

Which of the following sequences are graphical?

1. $s_1 : 3, 3, 2, 2, 1, 1$
2. $s_2 : 6, 5, 5, 4, 3, 3, 3, 2, 2$
3. $s_3 : 7, 6, 4, 4, 3, 3, 3$
4. $s_4 : 3, 3, 3, 1$

s_1 is a degree sequence of the graph,



Degree sequence

Theorem

A non-increasing sequence $s = \{d_1, d_2, \dots, d_n\}$ for $(n \geq 2)$ of nonnegative integers, where $d_1 \geq 1$, is graphical if and only if the sequence,

$$s_1 = \{d_2 - 1, d_3 - 1, \dots, d_{d_1+1} - 1, d_{d_1+2}, \dots, d_n\}$$

is graphical.

Sketchy Proof

1. Suppose, s_1 is graphical. So, there exists a graph G_1 such that,

$$\deg_{G_1}(v_i) = \begin{cases} d_i - 1 & 2 \leq i \leq d_1 + 1 \\ d_i & \text{otherwise} \end{cases}$$

2. Construct graph G from G_1 by adding a new vertex v_1 and d_1 edges $v_1 v_i$, for $2 \leq i \leq d_1 + 1$.
3. **Home work:** Prove the converse. Due: Next week, this class.

Example

Decide whether the sequence $s = \{5, 4, 3, 3, 2, 2, 2, 1, 1, 1\}$ is graphical or not.

1. Deleting 5 from s and subtracting 1 from the next five terms,

$$s'_1 = \{3, 2, 2, 1, 1, 2, 1, 1, 1\}$$

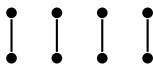
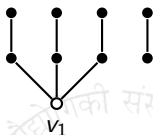
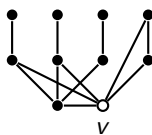
2. Reorder the sequence,

$$s_1 = \{3, 2, 2, 2, 1, 1, 1, 1, 1\}$$

3. s is graphical if and only if s_1 is. So, continue the procedure.
4. Deleting 3 from s_1 and subtracting 1 from the next three terms,

$$s_2 = \{1, 1, 1, 1, 1, 1, 1, 1\}$$

Getting the graph

 G_2  G_1  G

1. Sequence s_2 was obtained from s_1 by deleting 3 from s_1 and subtracting 1 from the next three terms of s_1 .
2. Add a new vertex v_1 to G_2 and join v_1 to three vertices of degree 1 in G_2 .
3. Sequence s_1 was obtained from s by deleting 5 from s and subtracting 1 from the next five terms of s .
4. Add vertex v to G_1 and join v to the vertex of degree 3, to two vertices of degree 2 and to two vertices of degree 1.



Paths and Circuits

Walk

Definition (Walk)

A *walk* is defined as a finite alternating sequence of vertices and edges, beginning and ending with vertices, such that each edge is incident with the vertices preceding and following it and no edge appears (is covered or traversed) more than once.

Definition

- Vertices with which a walk begins and ends, are called its *terminal vertices*.
- If a walk to begin and end at the same vertex, then it is called a *closed walk*. Otherwise, a walk is an *open walk*.
- An open walk in which no vertex appears more than once is called a *path*.

Proposition (Due: Next week, this class.)

Let G be a simple graph. Let w be a path in G . Prove that the edges of w are distinct.

Interesting ...

Definition

- Every walk expresses the idea of *reachability*.
- A vertex is said to be **reachable** from another vertex, if there exists a (open) walk such that starting a walk from later, one can eventually terminate the walk at the former.
- A path can not **intersect** itself.
- A self-loop can be a part of a walk but, not of a path.
- Every path in a graph, is a subgraph of the graph.
- Every vertex, except the starting and ending vertices, in a *path subgraph*, are of degree two and the terminal vertices are of degree one.

Definition (Trail)

A walk in which no edge appears more than once, is called a *trail*.

Some definition

Definition (Length)

The number of edges encountered in a walk (including multiple occurrences of an edge) is called the *length* of the walk.

Definition (Distance)

The *distance* between any two vertices u and v is the smallest length of any $u - v$ path in G denoted by $d_G(u, v)$ or $d(u, v)$. A $u - v$ path of length $d(u, v)$ is called a $u - v$ *geodesic*.

Definition (Diameter)

The greatest distance between any two vertices of a connected graph G is called the diameter of G and is denoted by $\text{diam}(G)$.

Concatenation and reversal

Proposition

Let G be a simple graph. Let u, v and w be three vertices of G . Let $a = (a_0, a_1, \dots, a_k)$ be a walk from u to v . Let $b = (b_0, b_1, \dots, b_l)$ be a walk from v to w . Then,

$$\begin{aligned}(a_0, a_1, \dots, a_k, b_0, b_1, \dots, b_l) &= (a_0, a_1, \dots, a_{k-1}, b_0, b_1, \dots, b_l) \\ &= (a_0, a_1, \dots, a_{k-1}, v, b_1, \dots, b_l)\end{aligned}$$

is a walk from u to w and it is denoted by $a * b$.

Proposition

Let G be a simple graph. Let u and v be two vertices of G . Let $a = (a_0, a_1, \dots, a_k)$ be a walk from u to v . Then,

1. The list $(a_k, a_{k-1}, \dots, a_0)$ is a walk from v to u . We denote this walk by **rev**(a) and call it the *reversal* of a .
2. If a is a path, then **rev**(a) is a path too.

A path is just a walk ...

Proposition

Let G be a simple graph. Let u and v be two vertices of G . Let $a = (a_0, a_1, \dots, a_k)$ be a walk from u to v . Assume that a is not a path. Then, there exists a walk from u to v whose length is smaller than some k .



A path is just a walk ...

Proposition

Let G be a simple graph. Let u and v be two vertices of G . Let $a = (a_0, a_1, \dots, a_k)$ be a walk from u to v . Assume that a is not a path. Then, there exists a walk from u to v whose length is smaller than some k .

Sketchy Proof

1. Since a is not a path, two of its vertices are equal.
2. So, there exist $i < j$ such that, $a_i = a_j$. Consider these i and j and the tuple,

$$\begin{array}{ccc} \text{B} & & \text{C} \\ @ & & \text{A} \\ \underbrace{a_0, a_1, \dots, a_i}_{\text{the first } i+1 \text{ vertices of } a} & , & \underbrace{a_{j+1}, a_{j+2}, \dots, a_k}_{\text{the last } k-j \text{ vertices of } a} \end{array}$$

3. It is a walk from u to v , and its length is $|\underbrace{\{z\}}_{< j}| + (k - j) < j + (k - j) = k$.

A different perspective

How do we answer below questions? Which one is stronger? Any algorithm?

1. Does G have a walk from u to v ?
2. Does G have a path from u to v ?
3. Find a shortest path from u to v (that is, a path from u to v having the smallest possible length), or determine that no such path exists.
4. Given a number $k \in \mathbb{N}$, find a walk from u to v having length k , or determine that no such walk exists.
5. Given a number $k \in \mathbb{N}$, find a path from u to v having length k , or determine that no such path exists.

Definition (Connectedness)

Let G be a simple graph. We define a binary relation, named **connectedness**, denoted by $\simeq_G$, on the set $V(G)$ as follows: For two vertices u and v of G , we shall have $u \simeq_G v$ if and only if there exists a walk from u to v in G .

Connectedness

Theorem

Let G be a graph of order 3 or more. If G contains two distinct vertices u and v such that $G - u$ and $G - v$ are connected, then G itself is connected.



Connectedness

Theorem

Let G be a graph of order 3 or more. If G contains two distinct vertices u and v such that $G - u$ and $G - v$ are connected, then G itself is connected.

Sketchy Proof

1. Suppose, x and y are two vertices of G . Consider two possible cases,
2. **Case I:** $(x, y) \neq (u, v)$. So, x and y are vertices in $G - u$ and $G - u$ is connected. Hence, there is a $x - y$ path, P , in $G - u$ which is present in G as well.
3. **Case II:** $(x, y) = (u, v)$ such that $x = u$ and $y = v$. Since G is of order 3 or more, there exists a vertex $w \neq u, v$ in G .
4. Since $G - v$ is connected, there must exist a $w - v$ path, P' . Also, $G - u$ is connected and so a $u - w$ path, P'' must exist in $G - u$.
5. Therefore, path P' followed by P'' ensures a $u - v$ path in G .

Connectedness

Theorem

Let G be a graph of order n . If

$$d(u) + d(v) \geq n - 1$$

for every two nonadjacent vertices u and v of G , then G is connected and $\text{diam}(G) \leq 2$. Also, the bound is **sharp**.



Connectedness

Theorem

Let G be a graph of order n . If

$$d(u) + d(v) \geq n - 1$$

for every two nonadjacent vertices u and v of G , then G is connected and $\text{diam}(G) \leq 2$. Also, the bound is **sharp**.

Sketchy Proof

1. Suppose, G is disconnected. Without affecting generality, also assume that n is even and G has two components G_1 and G_2 .
2. To maximize the degree, let G_1 and G_2 both have $\frac{n}{2}$ vertices and each component is completely connected within themselves.
3. Therefore, sum of degrees of u and v will be $\frac{n}{2} - 1 + \frac{n}{2} - 1 = n - 2$. This is the maximum sum value that can be obtained.
4. Hence, sum value of $(n - 2)$ does not necessarily guarantee connectedness.

Homework

Justify truth or falsity of the below statement

Every connected graph G of order 4 or more contains three distinct vertices u, v and w such that $G - u$, $G - v$ and $G - w$ are connected.

Generating regular graphs

Theorem

Let r and n be integers with $0 \leq r \leq n - 1$. There exists an r -regular graph of order n if and only if at least one of r and n is even.



Generating regular graphs

Theorem

Let r and n be integers with $0 \leq r \leq n-1$. There exists an r -regular graph of order n if and only if at least one of r and n is even.

Sketchy Proof

1. Showing the forward direction is easy as both r and n cannot be odd. **Why?**
2. So, let r and n be integers with $0 \leq r \leq n-1$ such that at least one of r and n is even. Suppose r is even and a graph $H_{r,n}$ with $V(H_{r,n}) = \{v_1, v_2, \dots, v_n\}$ is constructed.
3. So, $r = 2k \leq n-1$ for some nonnegative integer $k \leq \frac{n-1}{2}$.
4. For each i , ($1 \leq i \leq n$), join v_i with $v_{i+1}, v_{i+2}, \dots, v_{i+k}$ and also with $v_{i-1}, v_{i-2}, \dots, v_{i-k}$.
5. Since vertices are arranged cyclically (based on index), each vertex v_i is adjacent to the k vertices that immediately follow v_i and the k vertices that immediately precede v_i .

Generating regular graphs

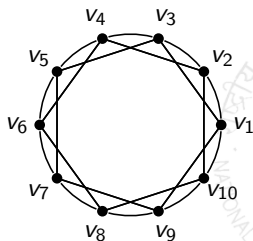
Sketchy Proof

6. Now suppose, n is even. So, $n = 2l$ and $r = 2k + 1 \leq \frac{n-1}{2}$ for some nonnegative integer $k \leq \frac{n-2}{2}$.
7. Join the the vertex v_i with $2k$ vertices as in the previous case for each i .
8. Further, join the vertex v_i with v_{i+l} .
9. Since vertices are arranged cyclically (based on index), each vertex v_i is adjacent to the k vertices that immediately follow v_i and the k vertices that immediately precede v_i . Also, to the vertex “opposite” to v_i .

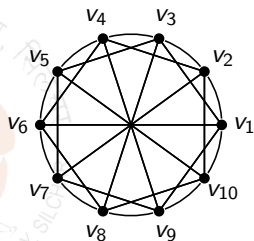
History

The graphs $H_{r,n}$ described above are called **Harary graphs**, named after Frank Harary.

An example



Harary Graph, $H_{4,10}$



Harary Graph, $H_{5,10}$

Generating regular graphs

Theorem

For every graph G and every integer $r \geq \Delta(G)$, there exists an r -regular graph H containing G as an induced subgraph.



Generating regular graphs

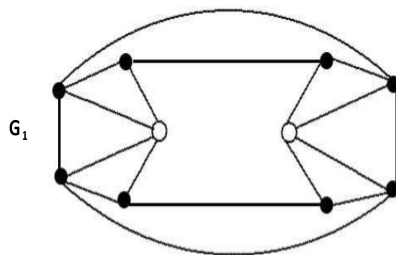
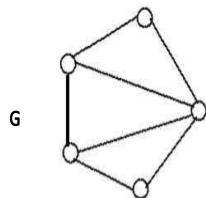
Theorem

For every graph G and every integer $r \geq \Delta(G)$, there exists an r -regular graph H containing G as an induced subgraph.

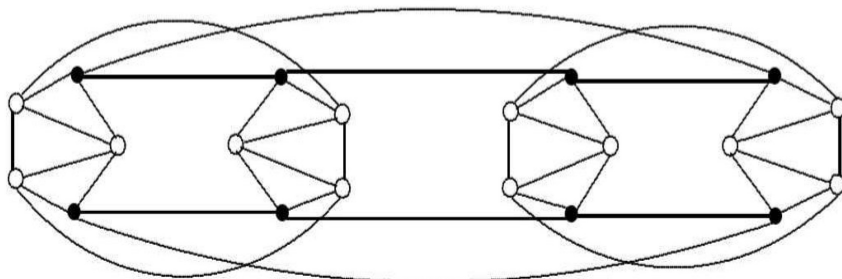
Sketchy Proof

1. If G is r -regular, then let $H = G$. Assume that G is not an r -regular graph. Suppose, G has order n and $V(G) = \{v_1, v_2, \dots, v_n\}$.
2. Let G' be another copy of G with $V(G') = \{v'_1, v'_2, \dots, v'_n\}$, where each vertex v'_i in G' corresponds to v_i in G for $1 \leq i \leq n$.
3. Construct a graph G_1 from G and G' by adding the edges $v_i v'_i$ for all vertices v_i ($1 \leq i \leq n$) of G for which $d(v_i) < r$.
4. G is an induced subgraph of G_1 and $\delta(G_1) = \delta(G) + 1$. If G_1 is r -regular, let $H = G_1$.
5. Else, repeat until arriving at an r -regular graph G_k , where $k = r - \delta(G)$.

An example



H



Homework

Show that

For a graph G of order n such that, $d(u) + d(v) \geq (n - 2)$, for every pair (u, v) of nonadjacent vertices of G , G has at most two components. Is the bound sharp?

Some more definition

Definition (Cycle)

A closed walk, $(v_1, v_2, \dots, v_k)$, such that, $v_1 = v_k$, is also called **circuit**. If in addition, $k \geq 3$ and each of the vertices in $\{v_1, v_2, \dots, v_{k-1}\}$, are distinct, then $(v_1, v_2, \dots, v_k)$ is called a *cycle*.

Definition (Grith)

The length of the shortest cycle contained in the graph is called *grith*.

Definition (Component)

A disconnected graph consists of two or more connected graphs. Each of these connected subgraphs is called a *component*. It is generally denoted by κ .

Theorem

A graph G is disconnected if and only if its vertex set V can be partitioned into two nonempty, disjoint subsets, V_1 and V_2 such that, there exists no edge in G whose one end vertex is in subset V_1 and the other in subset V_2 .

Edges in disconnected graph

Theorem

A simple graph, G , with n vertices and k components can have at most $(n - k)(n - k + 1)/2$ edges.



Edges in disconnected graph

Theorem

A simple graph, G , with n vertices and k components can have at most $(n - k)(n - k + 1)/2$ edges.

Sketchy Proof

- Let $n_1, n_2, \dots, n_k$ be the number of vertices in each components of G .
Therefore,

$$n_1 + n_2 + \dots + n_k = n$$

- Also, $\sum_{i=1}^k (n_i - 1) = n - k$. So, squaring both sides and some algebraic manipulation leads to,

$$\sum_{i=1}^k n_i^2 \leq n^2 - (k - 1)(2n - k)$$

- Maximum number of edges possible in the i^{th} component of G is $\frac{1}{2}n_i(n_i - 1)$.

Sum it.

Odd degree and path

Theorem

If a graph (connected or disconnected) has exactly two vertices of odd degree, there must be a path joining these two vertices.



Odd degree and path

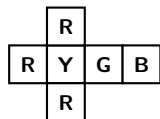
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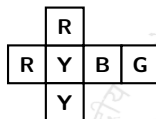
Sketchy Proof

1. Suppose v_1 and v_2 are exactly two vertices of odd degree in the graph G .
2. If G is connected, it is trivial.
3. Otherwise, if v_1 and v_2 happen to belong to two different components, there does not exist any path between them.
4. So, both the components will have exactly one vertex of odd degree.
5. Two subgraphs are obtained with odd number of odd degree vertices. **Impossible. Why?**

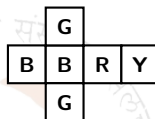
A puzzle



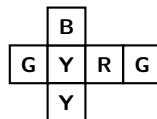
Cube 1



Cube 2



Cube 3



Cube 4

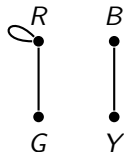
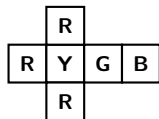
The puzzle

Given four cubes whose faces are colored red, blue, green and yellow, can we pile them up so that all four colors appear on each side of the resulting 4×1 stack?

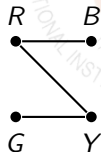
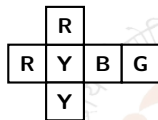
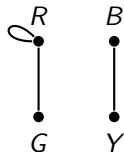
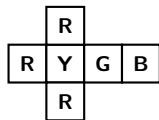
The steps

1. Represent each cube by a graph with four vertices $\{R, B, G, Y\}$, one for each color.
2. Two vertices are adjacent if and only if the cube in question has the corresponding colors on the **opposite faces**.
3. Take **union** of all these graphs to generate one graph G .
4. Find two subgraphs G_1 and G_2 of G such that,
 - a) each subgraph contains exactly one edge from each cube.
 - b) the subgraphs have no edges in common.
 - c) each subgraph is 2-regular.

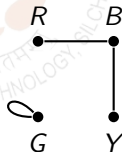
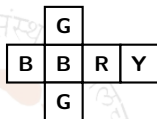
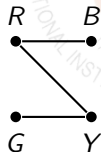
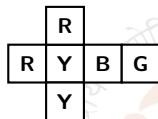
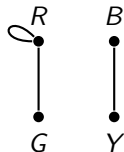
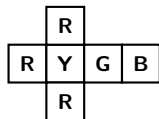
Obtaining the graphs



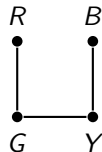
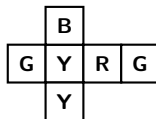
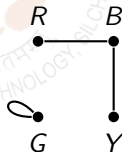
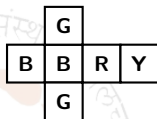
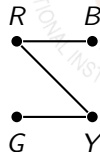
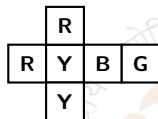
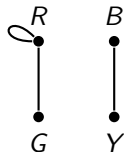
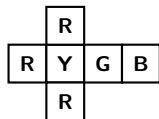
Obtaining the graphs



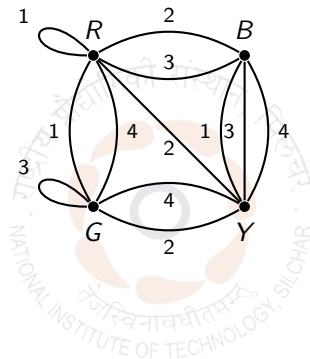
Obtaining the graphs



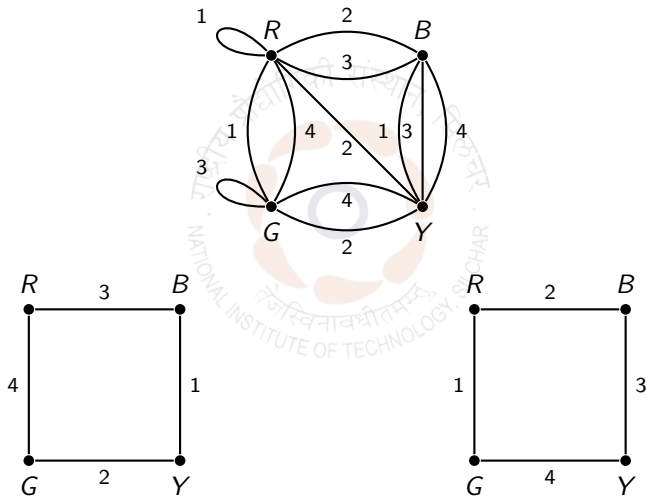
Obtaining the graphs



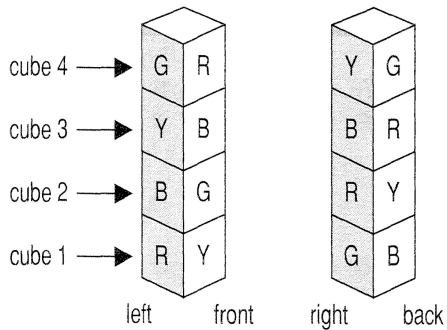
Union and the subgraphs



Union and the subgraphs



The solution



Homework

Show that

A graph G has odd order if $G - v$ is 3-regular, where v is a vertex of G .

The Expanding Unicurse

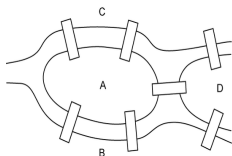
Some citizens of Königsberg
Were walking on the strand
Beside the river Pregel
With its seven bridges spanned.
“O Euler, come and walk with us,”
Those burghers did beseech.
“We’ll roam the seven bridges o’er,
And pass but once by each.”
“It can’t be done,” thus Euler cried.
“Here comes the Q. E. D.
Your islands are but vertices,
And four have odd degree.”
From Königsberg to König’s book,
So runs the graphic tale,
And still it grows more colorful,
In Michigan and Yale.

William T. Tutte (1917-2002)

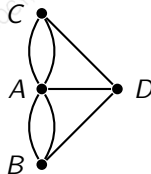
Eular graph

History

The concept of Eular graph arises out of famous problem of Königsberg Bridge problem. The river Pregel running in the city of Königsberg (later renamed as Kalinigrad) in East Prussia, had two river islands connected to main land by seven bridges. The question arouse, *if any person, starting from any of the land mass, can traverse each of the bridges exactly once and yet return to the starting location?* **Leonhard Eular**, with the help of graph theory, solved this problem in 1736 and the paper published by him in this regard, is considered to be the foundation of the field of graph theory.



(a) A pictorial representation



(b) A graph representation

Eular graph I

Definition (Eular graph)

A graph having a closed walk such that every edge of the graph is traversed (covered or visited) exactly once, is known as *Eular graph* and the corresponding walk is called *closed Eular walk* or *Eular circuit*

Remark

An Eular graph has to be a connected graph. However, an Eular graph can have isolated vertices.

Theorem

A connected graph G is an Eular graph if and only if every vertex of G are of even degree.

Eular graph II

Sketchy Proof

1. Suppose G is an Eular graph and consider an arbitrary vertex v on the Eular trail.
2. Use an edge e of V to reach vertex u (say). Mark e as "exit" for v and "entry" for u . Repeat the process with u and so on till reaching v again.
3. Every vertex, so visited, must have at least a pair of (entry, exit) edges.
4. Conversely, start with an arbitrary vertex v . Traverse as many vertices as possible so that the walk ends in vertex v , without traversing any edge more than once.
5. Possible, because, for every exit edge from a vertex, there must exist an entry edge into the vertex.

Eular graph

Theorem

Let $G = (V, E)$ be a graph and suppose that t is a nontrivial tour (closed trail). Then, t is composed of edge-disjoint cycles.



Eular graph

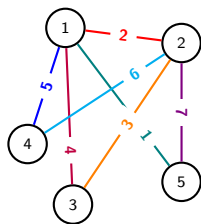
Theorem

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Sketchy Proof

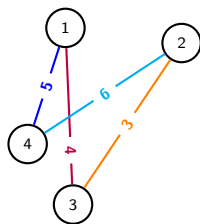
1. Proof by induction on length of the tour, l .
2. Say, t is a **one-edge closed tour**, then G is a non-simple graph that contains a self-loop.
3. Suppose the result holds for all closed trails of length n or less.
4. Say, $l = n + 1$ and there is at least one cycle c in t . Remove c from this tour to obtain a new tour t' . The tour t' has a length of at most n .

An example



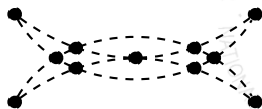
$$=$$

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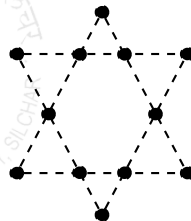


Fun side

Draw the following figures in one continuous line without retracing and without lifting the pencil from the paper.



Mohammed's scimitar



Star of David

Homework

Comment and justify

A graph G obtained by union of two vertex disjoint graphs G_1 and G_2 such that, for every $v_1 \in V(G_1)$ and $v_2 \in V(G_2)$, $v_1 v_2 \in E(G)$.

Remark

This operation is also called **join** of graphs G_1 and G_2 and it is denoted by, $G = G_1 + G_2$.

Eular line

Definition (Unicursal graph)

A graph in which a walk, not necessarily a closed one, is found such that, each edge is traversed exactly once, is called *unicursal graph*. Such a walk is called *unicursal line* or *open Eular line* or *Eular trail*.

Remark

Surely, if an edge is dropped from an Euler circuit, a unicursal line is obtained. However, a unicursal graph is not necessarily an Euler graph but, adding an edge between starting and ending vertices of a unicursal line results into an Euler graph.

Theorem

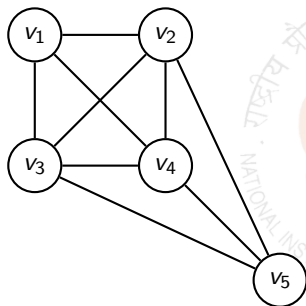
A connected graph G is a unicursal graph if and only if G has exactly two vertices of odd degree. Furthermore, each Eulerian trail of G begins at one of these odd vertices and ends at the other.

Eular line

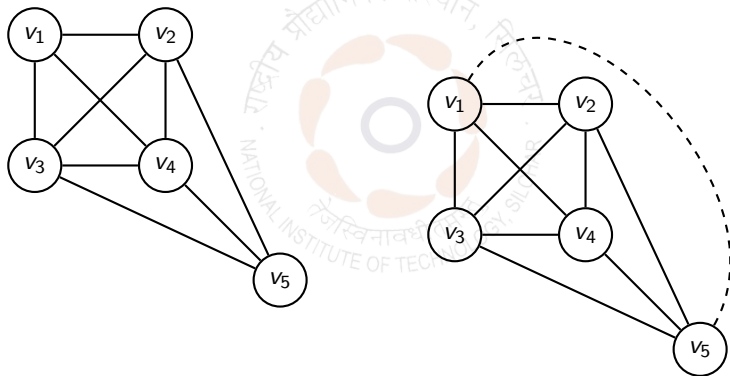
Sketchy Proof

1. Suppose $u - v$ is an Eularian trail in the graph G .
2. Construct a new graph, H , by adding a new vertex x to G and connecting both u and v to x by an edge. So, H is an Eular graph. **Little more argument.**
3. Showing the converse is similar.
4. Construction of H shows that trail has to begin either at u or v (and necessarily end at v or u , respectively). Consideration of any other vertex will violate the Eular property.

An example



An example



Finding Eular cycle

Fleury's Algorithm:

Input: An Eulerian graph G .

$W \leftarrow \emptyset; H \leftarrow G$

Pick any vertex, v , as *lead*.

$W \leftarrow W \cup \{v\}$

while (True) **do**

if All edges of H have been deleted **then**

Return W

else

Continue

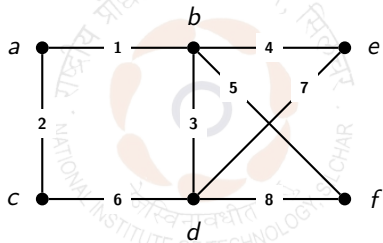
Use an edge, e , incident on the *lead* vertex, to travel to adjacent vertex, u , only if deleting that edge would not disconnect the graph H (isolated vertex ignored).

Set u as *lead*.

$W \leftarrow W \cup \{e\} \cup \{u\}$

$H \leftarrow H - \{e\}$.

Try



Fleury's Algorithm

Theorem

If G is an Euler graph, the walk W returned by Fleury's Algorithm is an Euler tour of G .



Fleury's Algorithm

Theorem

If G is an Euler graph, the walk W returned by Fleury's Algorithm is an Euler tour of G .

Sketchy Proof

1. First, show that W is a closed trail of G . The algorithm always deletes an edge of G which is incident on to the terminal vertex of W .
2. The algorithm terminates when all the edges incident on to the terminal vertex of W have been deleted.
3. Since G is even, terminal vertex = v . So, closed.
4. To show that W is an Euler tour, proceed by *contradiction*. Suppose, X is the set of vertices of positive degree in H when the algorithm terminates.
5. What happens, if $G(X)$ is an even subgraph of G ? Otherwise?

A generalization

Theorem

A connected graph G with exactly $2k$ odd degree vertices, have exactly k edge disjoint subgraphs such that, they together contain all the edges of G and each subgraph is a unicursal graph.



A generalization

Theorem

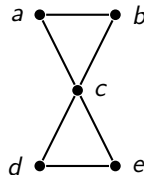
A connected graph G with exactly $2k$ odd degree vertices, have exactly k edge disjoint subgraphs such that, they together contain all the edges of G and each subgraph is a unicursal graph.

Sketchy Proof

1. Say, the odd vertices are $v_1, v_2, \dots, v_k, w_1, w_2, \dots, w_k$.
2. Add k edges between the pair of vertices $(v_1, w_1), (v_2, w_2), \dots, (v_k, w_k)$.
3. So, G is Eulerian with Euler tour ρ .
4. Remove from ρ each of the k edges added one-by-one.
5. ρ is split into k walks, each is a unicursal line.

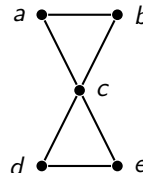
Little more on Eular graph

- Start Euler tour on the Euler graph as $a \rightarrow b \rightarrow c$.
- Next, choosing $c \rightarrow a$ edge precludes finding the tour.
- What property must a vertex v possess such that an Euler line is always obtained when one follows any walk from vertex v ?



Little more on Euler graph

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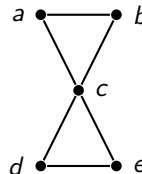


Definition (Arbitrarily traceable graph)

If an Euler tour is always found starting at vertex v , irrespective of which edge has been chosen, such a graph is called *arbitrarily traceable graph from v* .

Little more on Euler graph

- Start Euler tour on the Euler graph as $a \rightarrow b \rightarrow c$.
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Definition (Arbitrarily traceable graph)

If an Euler tour is always found starting at vertex v , irrespective of which edge has been chosen, such a graph is called *arbitrarily traceable graph from v* .

Theorem

An Euler graph G is arbitrarily traceable from vertex v if and only if every cycle in G contains v .

Homework

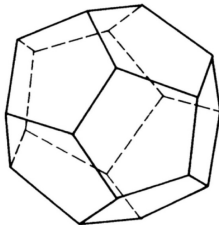
Give an example of a graph G such that,

- a) both G and $\overline{G}$ are Eulerian.
- b) G is Eulerian but $\overline{G}$ is not.
- c) neither G nor $\overline{G}$ is Eulerian and both G and $\overline{G}$ contain an Eulerian trail.

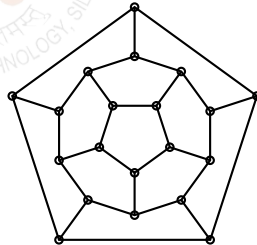
Hamiltonian cycle

History

A regular dodecahedron has been known to be used as a desk calendar since a month can be displayed on each of its twelve faces. In 1857, an Irish mathematician, Sir William Rowan Hamilton invented a game called the *Icosian Game* and patented in London in 1859. It consisted of a regular dodecahedron made from wood. Each corner of the dodecahedron was marked with one of twenty cities beginning with different consonants. A goal of the game was to travel “Around the World,” that is, to *find a round trip that passed through each of the twenty cities exactly once*.



(a) Dodecahedron



(b) A graph representation

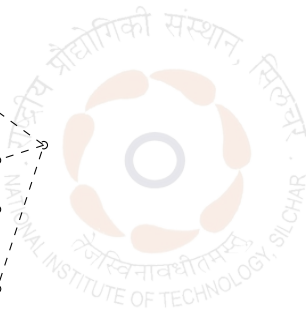
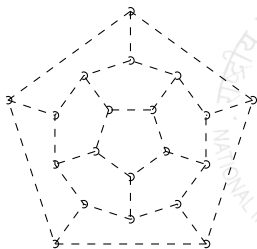
Hamiltonian cycle

Definition (Hamiltonian cycle)

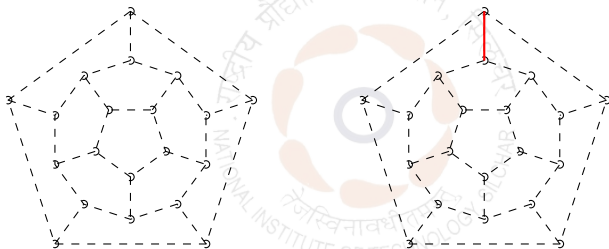
A circuit in a graph G is said to be *Hamiltonian* if it includes every vertex of G . A *Hamiltonian path* in G means a walk of G that contains each vertex of G exactly once.

- Having a Hamiltonian cycle is stronger than having a Hamiltonian path, because cycle includes path but not vice-versa.
- If G is a Hamiltonian graph, then certainly G is connected.
- No single vertex removal can disconnect G .

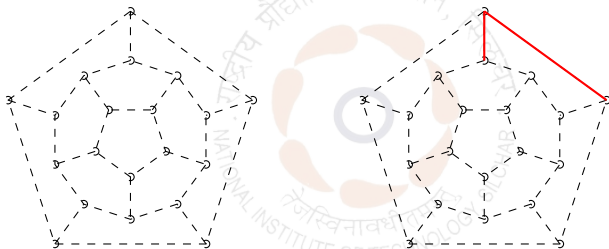
An example



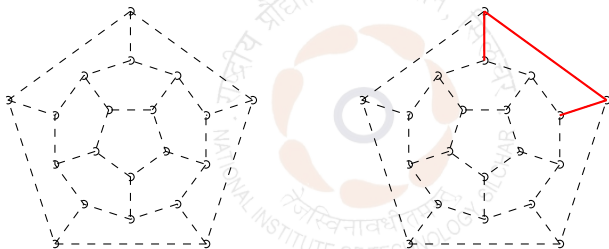
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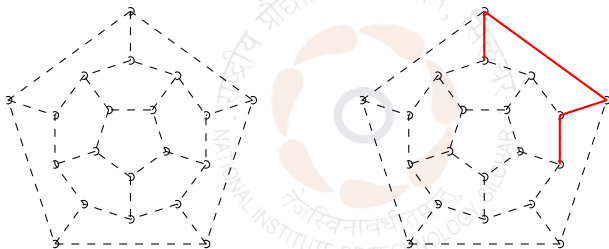
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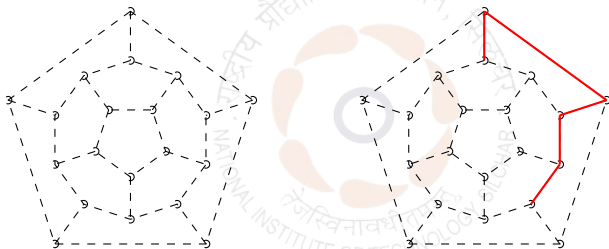
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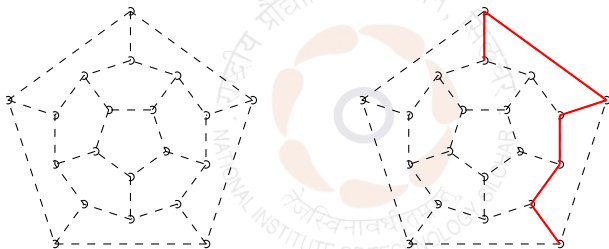
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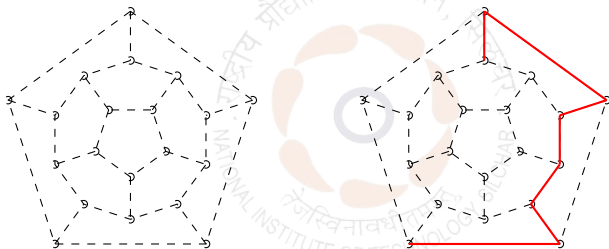
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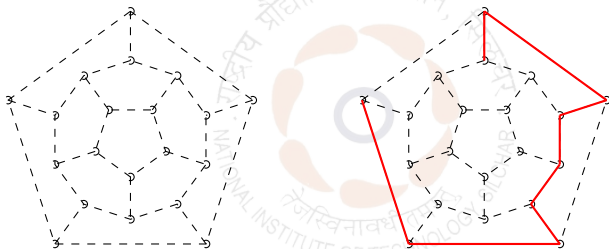
An example



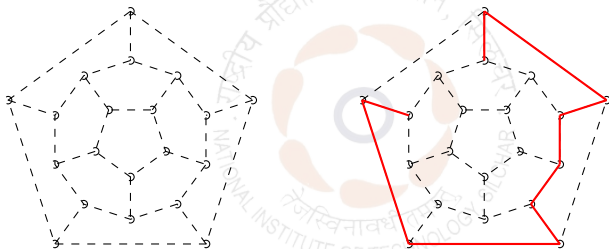
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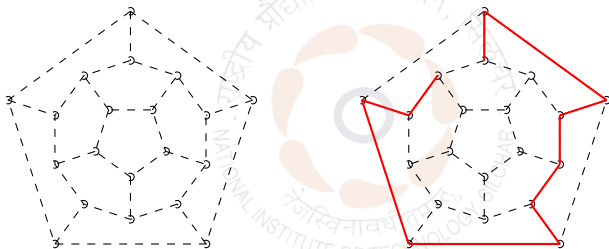
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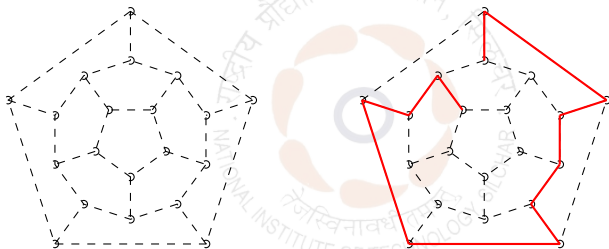
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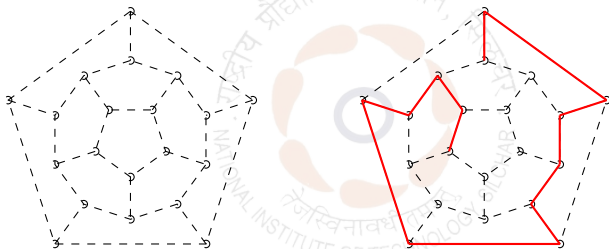
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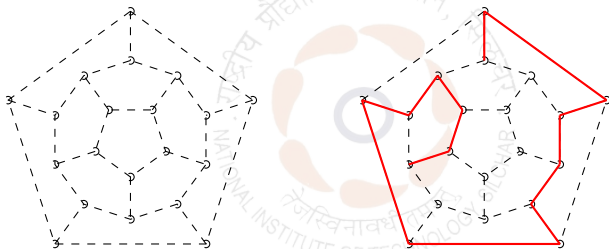
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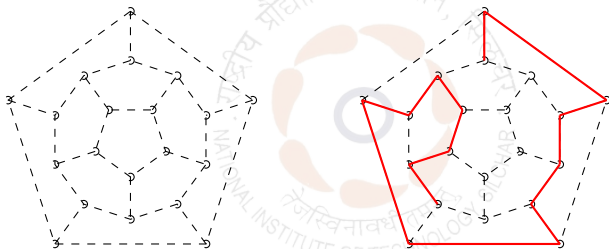
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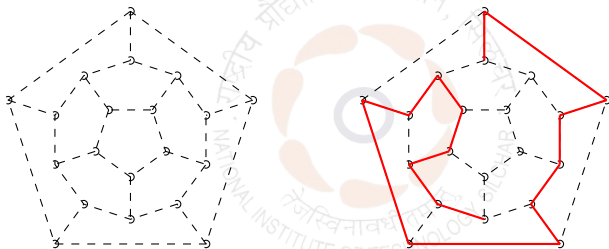
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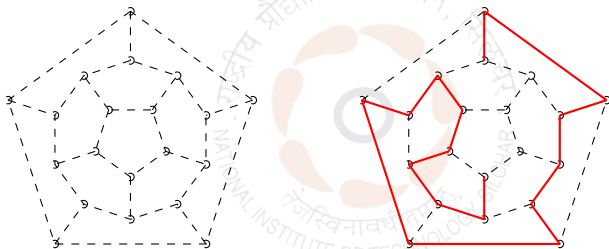
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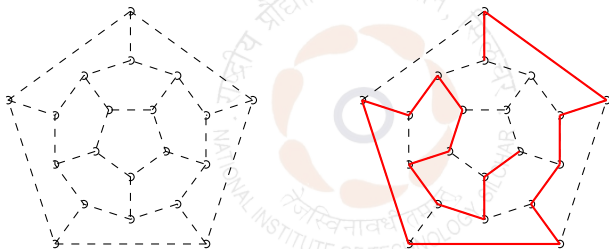
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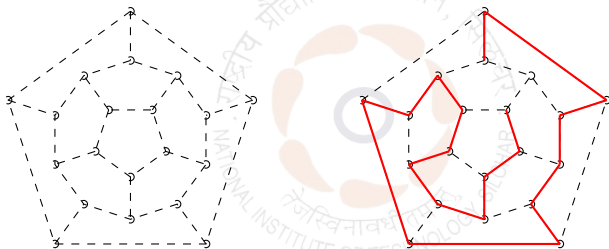
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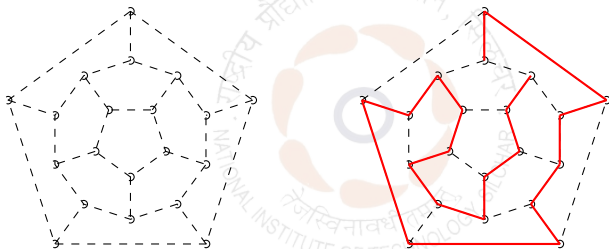
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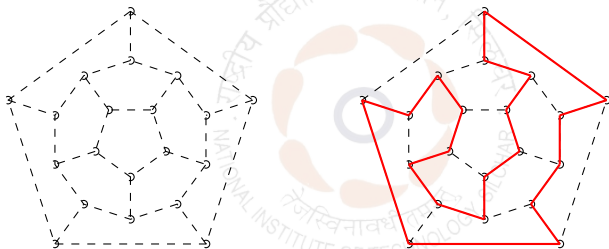
An example



An example



An example



Hamiltonian cycle

Lemma

If G is a Hamiltonian graph, then for every nonempty proper set S of vertices of G ,

$$\kappa(G - S) \leq |S|$$

Sketchy Proof

1. Say, $S \subset V(G)$ and $\kappa(G - S) = k$.
2. $G_1, G_2, \dots, G_k$ are the components of $G - S$.
3. Since G is Hamiltonian, it contains a Hamiltonian cycle C .
4. Whenever C encounters a vertex of G_i **for the last time** ($1 \leq i \leq k$), the next vertex of C must belong to S . Otherwise, it is not the last time.
5. So, S must contain at least k vertices or, $\kappa(G - S) \leq |S|$

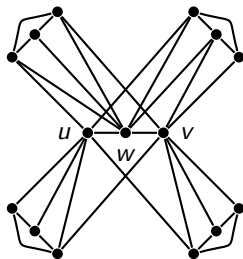
Hamiltonian cycle

- The main benefit of the necessary condition is the statement,

Contrapositive

Let G be a graph. If $\kappa(G - S) > |S|$ for some nonempty proper subset S of $V(G)$, then G is not Hamiltonian.

- This gives a sufficient condition for a graph to be non-Hamiltonian.



- $S = \{u, w, v\}$

- $\kappa(G - S) = 4 > 3 = |S|$

- Non-Hamiltonian.



Hamiltonian cycle

Theorem

Let G be a graph of order $n \geq 3$. If

$$d(u) + d(v) \geq n$$

for each pair u, v of nonadjacent vertices of G , then G is Hamiltonian.

Sketchy Proof

1. Assume, there exists such non-Hamiltonian graph, G .
2. Add as many edges as possible to G so that the resulting graph H is not Hamiltonian such that, adding any edge to H results in a Hamiltonian graph.
3. Since H is not complete, H contains pairs of nonadjacent vertices, say, x and y .
4. Thus, H contains a Hamiltonian $x - y$ path $P = (x = x_1, x_2, \dots, x_n = y)$.

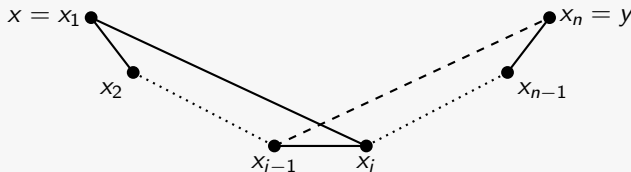
Hamiltonian cycle

Sketchy Proof

5. So, whenever (x_1, x_i) is an edge of H , where $2 \leq i \leq n$, then (x_{i-1}, x_n) is not an edge of H . Otherwise,

$$(x_1, x_i, x_{i+1}, \dots, x_n, x_{i-1}, x_{i-2}, \dots, x_1)$$

is a Hamiltonian cycle of H . **Impossible.**



6. Hence, for each vertex in $\{x_2, x_3, \dots, x_n\}$ that is adjacent to x_1 , there is a vertex in $\{x_1, x_2, \dots, x_{n-1}\}$ that is not adjacent to x_n .
7. Therefore, $d(x_n) \leq (n-1) - d(x_1)$. **Contradiction.**

Hamiltonian cycle

Theorem

A graph is Hamiltonian if and only if its closure is Hamiltonian.

Sketchy Proof

1. If a graph, G of order n , is Hamiltonian, surely, $C(G)$ will be Hamiltonian too.
2. Conversely, assume $C(G)$ is Hamiltonian but G is not.
3. Suppose, at the k^{th} step $C(G)$ is obtained i.e. $C^k(G) = C(G)$.
4. Say, Hamiltonian cycle, Y , is present in $C(G)$ which is obtained by adding the edge (x, y) to $C^{k-1}(G)$.
5. Therefore,

$$d_{C^{k-1}(G)}(x) + d_{C^{k-1}(G)}(y) \geq n$$

Hamiltonian cycle

Theorem

Let G be a graph of order $n \geq 3$. If for every integer j with $1 \leq j < \frac{n}{2}$, the number of vertices of G with degree at most j is less than j , then G is Hamiltonian.

Sketchy Proof

1. Suppose, $C(G)$ is not complete but Hamiltonian. Let (u, v) be a pair of non-adjacent vertices satisfying maximum value of $d_{C(G)}(u) + d_{C(G)}(v)$ among all such nonadjacent pairs. So,

$$d_{C(G)}(u) + d_{C(G)}(v) \leq n - 1$$

2. Assume that, $d_{C(G)}(u) \leq d_{C(G)}(v)$ and, say, $d_{C(G)}(u) = k$. Therefore, $k \leq \frac{n-1}{2}$ and hence,

$$d_{C(G)}(v) \leq n - k - 1$$

3. Let V be the set of all vertices distinct from v that are not adjacent to v . Surely, $u \in V$.

Hamiltonian cycle

Sketchy Proof

4. If some $x \in V$, then $d_{C(G)}(x) \leq k$, otherwise,

$$d_{C(G)}(x) + d_{C(G)}(v) > d_{C(G)}(u) + d_{C(G)}(v)$$

a contradiction to the choice of (u, v) pair.

5. Therefore, degree of every vertex in V is at most k .

6. So, $|V| \leq k$. However, $|V| = k$ cannot produce a Hamiltonian cycle. Hence, $|V| \leq k - 1$. This implies,

$$d_{C(G)}(v) \geq (n - 1) - (k - 1) = n - k$$

7. A contradiction.

Hamiltonian cycle

Observation

For $j = 1$, G has no vertex of degree 1. For $j = 2$, the graph G is allowed to have a vertex of degree 2. For $j = 3$, the graph G is allowed to have a vertex of degree 2 and a vertex of degree 3 or two vertices of degree 3 and so on.

On counting Hamiltonian cycles

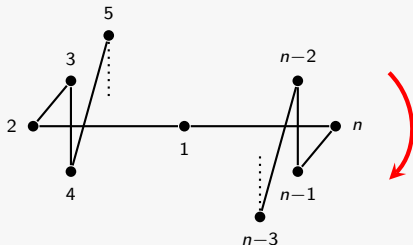
Theorem

In a complete graph with n vertices there are $(n-1)/2$ edge-disjoint Hamiltonian cycles, if n is an odd number ≥ 3 .

Sketchy Proof

1. A complete graph G of n vertices has $\frac{n(n-1)}{2}$ edges and a Hamiltonian cycle in G consists of n edges.
2. Therefore, the number of edge-disjoint Hamiltonian cycles in G cannot exceed $\frac{n(n-1)}{2}$.

The subgraph of the complete graph of n vertices, is a Hamiltonian cycle.



On counting Hamiltonian cycles

Sketchy Proof

3. Keep the vertices fixed on a circle and rotate the polygon in clockwise manner, by $\frac{360}{n-1}, 2 \cdot \frac{360}{n-1}, \dots, \frac{(n-3)}{2} \cdot \frac{360}{n-1}$ degrees.
4. Each rotation produces a new Hamiltonian cycle that has no edge in common with any of the previous ones.
5. So, $\frac{n-3}{2}$ new Hamiltonian cycles are generated, each of which is edge disjoint with every other, including the one in the figure.

Homework

Comment and justify

Let G be a 3-regular graph of order 12 and H a 4-regular graph of order 11.

- a) Is $G + H$ Eulerian?
- b) Is $G + H$ Hamiltonian?

Introduction

Definition (Metric)

A function $f(x, y)$ of two variables can be called *metric* if it satisfy the following,

1. **Non-negativity:** $f(x, y) \geq 0$ and $f(x, y) = 0$ iff $x = y$.
2. **Symmetry:** $f(x, y) = f(y, x)$.
3. **Triangular inequality:** $f(x, y) \leq f(x, z) + f(z, y), \forall z$.

Theorem

Distance between vertices in a connected graph is a metric.

Distance as a metric

Sketchy Proof

1. For any two vertices $u, v \in V(G)$, the distance, $d(u, v)$, can either be 0 or some non-negative value.
2. In an undirected graph, the number of edges traversed during the walk from u to v will be the same as the number of edges traversed during the shortest walk from v to u . (This is, in general, not true for **directed** graph.)
3. Say, $w \in V(G)$ and $w \neq u, v$. If walk from u to v passes through w , $d(u, v) = d(u, w) + d(w, v)$. Otherwise, any excursion from u to w and then to v must be a longer walk.

Some more

Definition (Eccentricity)

Eccentricity of a vertex in a graph, denoted by $\mathcal{E}(v)$, is the distance of the vertex v to the farthest vertex in the graph G i.e.

$$\mathcal{E}(v) = \max_{v_i \in V(G)} d(v, v_i)$$

Definition (Center)

A vertex with minimum eccentricity is known as *center* of a graph.

Definition (Radius)

The eccentricity value of the center of a graph is called *radius* of the graph and it is denoted by, $\text{rad}(G)$ for graph G .

Yet more

Remark

A graph can have many centers. For example, every vertex of a circuit is a center.

Definition (Degree centrality)

The *degree centrality* of a vertex is just its degree. It is the simplest measure of centrality.

Definition (Geodesic centrality)

The *geodesic centrality* (also called the *betweenness centrality*) of a vertex $v \in V$, is the fraction of times v occurs on any shortest path connecting any other pair of vertices $s, t \in V$.

Remark

There are other measures of centrality such as, *Closeness centrality*, *Eigenvector centrality*, *Page rank centrality* etc.

A result

Theorem

If G is a simple graph, then $\text{diam}(G) \geq 3 \Rightarrow \text{diam}(\overline{G}) \leq 3$.



A result

Theorem

If G is a simple graph, then $\text{diam}(G) \geq 3 \Rightarrow \text{diam}(\overline{G}) \leq 3$.

Sketchy Proof

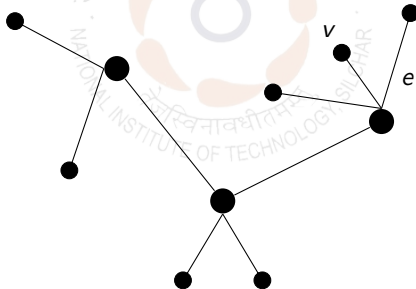
1. If $\text{diam}(G) > 2$, there must exist two non-adjacent vertices u and v such that they have no common neighbor.
2. So, every vertex $x \in V(G) - \{u, v\}$ must have at least one of u or v as non-neighbor.
3. Therefore, x is adjacent to either u or v in $\overline{G}$. Also, there will be an edge between u and v in $\overline{G}$.
4. Hence, for every x, y pair of vertices, there will be a $x - y$ path through uv edge.
5. Further, length of the path will be no longer than 3.



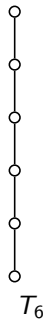
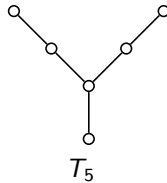
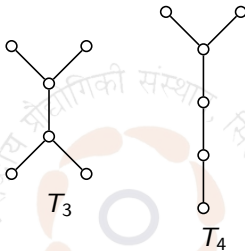
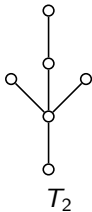
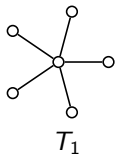
Tree

Definition (Tree)

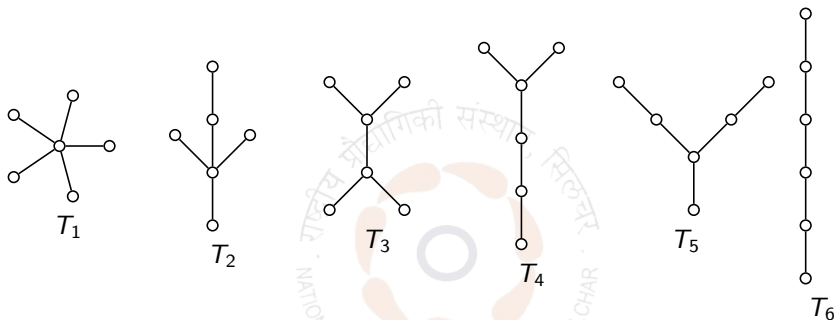
A *tree* is a connected graph without any circuit. In other words, a tree is an acyclic connected graph.



Trees of order 6



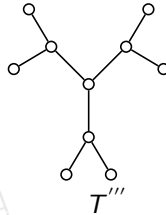
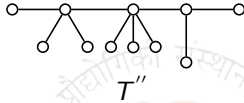
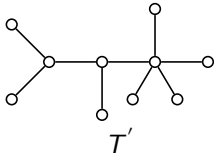
Trees of order 6



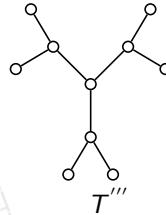
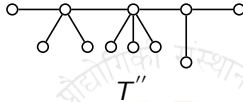
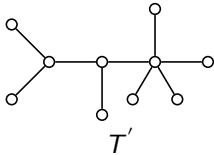
Remark

1. The tree $T_1 = K_{1,5}$ is a *star* and $T_6 = P_6$ is a *path*.
2. A tree containing exactly two vertices that are not end vertices (which are necessarily adjacent) is called a **double star**. T_2 and T_3 , for example.
3. A **caterpillar** is a tree of order 3 or more, the removal of whose end-vertices produces a path called the **spine** of the caterpillar.

Caterpillar



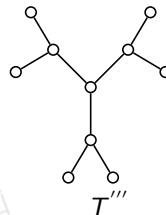
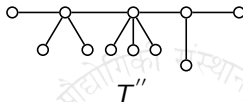
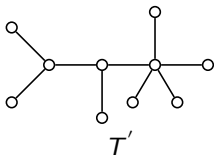
Caterpillar



Remark

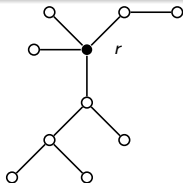
1. Every path, star (of order at least 3) and every double star is a caterpillar.
2. T' and T'' are caterpillars but T''' is not.

Caterpillar



Remark

1. Every path, star (of order at least 3) and every double star is a caterpillar.
2. T' and T'' are caterpillars but T''' is not.



A rooted tree

Definition (Rooted tree)

Sometimes it is convenient to select a vertex of a tree T under discussion and designate this vertex as the **root** of T . The tree T then becomes a **rooted tree**.

Some results

Theorem

A connected graph G is a tree if and only if there exists one and only one path between every pair of vertices.



Some results

Theorem

A connected graph G is a tree if and only if there exists one and only one path between every pair of vertices.

Sketchy Proof

1. Say, G is a tree with n vertices and p_{uv} & q_{uv} are two distinct paths between any two arbitrary vertices u and v of G .
2. If $q_{uv} \neq p_{uv}$, there will be a circuit in G .
3. Conversely, suppose there is only one path between any pair of vertices u and v .
4. Then this path will have to be traversed while walking from v to u .
5. Thus no circuit can be formed.

Some results

Theorem

A connected simple graph G of n vertices has $n - 1$ edges if and only if it is a tree.



Some results

Theorem

A connected simple graph G of n vertices has $n - 1$ edges if and only if it is a tree.

Sketchy Proof

1. Prove by induction on number of vertices assuming G is of size $n - 1$.
2. **Basis:** Consider $n = 1$. one isolated vertex and there can be no edge.
3. **Hypothesis:** Assume that the statement is true for $n = k$, $k < n$.
4. **Induction:** Suppose $n = k + 1$ and so, one new edge e_k is introduced in the graph.
5. If e_k forms a circuit in the graph, then end-vertices of e_k must belong to already existing vertices ($\neq v_{k+1}$) of the graph.
6. In such a case v_{k+1} is an isolated vertex. **Complete the converse.**

Some results

Definition (Minimally connected graph)

A graph is said to be *minimally connected* if removal any *one edge* from it, renders the graph to be disconnected.



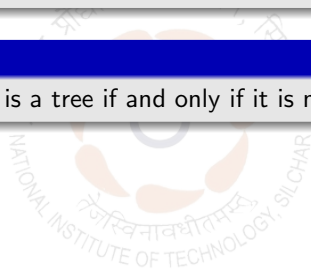
Some results

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A connected simple graph is a tree if and only if it is minimally connected.



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Lemma

Every tree with at least two vertices has at least two leaves. Deleting a leaf vertex from a tree of n vertices produces a tree of $(n - 1)$ vertices.

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Lemma

Every tree with at least two vertices has at least two leaves. Deleting a leaf vertex from a tree of n vertices produces a tree of $(n - 1)$ vertices.

Sketchy Proof

1. Suppose v is a leaf vertex in the tree T of n vertices. Let, $T' = T - \{v\}$.
2. For any $x, y \neq v$, p_{xy} exists in T' . So, T' is connected. Rest?

Homework

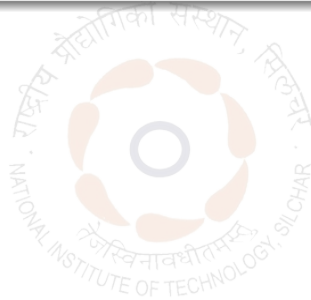
Lemma

Given a tree T of n vertices, $n \geq 4$ and a vertex v of maximum degree in T , show that, T is a path if $d(v) = 2$.

Forest

Definition (Forest)

A *forest* is an acyclic graph. Forests consist only of (possibly disconnected) trees.



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Sketchy Proof

1. Say, size of a forest F is m . Let $G_1, G_2, \dots, G_k$ be the components of F .
2. Also, suppose G_i has order n_i and size m_i for $1 \leq i \leq k$. Therefore, $n = \sum_{i=1}^k n_i$ and $m = \sum_{i=1}^k m_i$.
3. Each of G_i is a tree. So, $m_i = n_i - 1$. Hence,

$$m = \sum_{i=1}^k m_i = \sum_{i=1}^k (n_i - 1) = n - k$$

Some more

Theorem

Every tree either has a vertex as its center or an edge as its center.



Some more

Theorem

Every tree either has a vertex as its center or an edge as its center.

Sketchy Proof

1. Prove by induction on number of vertices.
2. **Basis:** Consider $n = 1$. This means there is only one vertex in the tree without any edge. Therefore, this vertex must be the center.
3. **Hypothesis:** Assume that the statement is true for $n = k$, $k < n$.
4. **Induction:** Suppose $n = k + 1$. So, the $(k + 1)^{\text{th}}$ vertex will increase the number of edges by one.
5. If a vertex, u , was the center, then one of its neighbor v , will have the same eccentricity value now, as that of u . So, uv edge is the new center. Reverse argument holds if uv edge was the center earlier.

Some more

Theorem

Let T be a tree of order n . If G is a graph with $\delta(G) > n - 1$, then T is isomorphic to some subgraph of G .



Some more

Theorem

Let T be a tree of order n . If G is a graph with $\delta(G) > n - 1$, then T is isomorphic to some subgraph of G .

Sketchy Proof

1. **Basis:** The result is trivially true for $n = 1$; every graph contains a vertex.
2. **Hypothesis:** Assume for every tree T' of order $k - 1$, where $k \geq 3$, and for every graph H with minimum degree at least $k - 2$ that T' is isomorphic to a subgraph of H .
3. **Induction:** Suppose $n = k$ and $\delta(G) > k - 1$.
4. Let v be an end-vertex of T and u is the vertex of T adjacent to v . Then $T - v$ is a tree of order $k - 1$.

Some more

Sketchy Proof

5. Since $\delta(G) \geq k - 1 > k - 2$, it follows by the induction hypothesis that $T - v$ is isomorphic to a subgraph F of G . Let u' denote the vertex of F corresponding to u in T .
6. Since $d_G(u') \geq k - 1$ and the order of F is $k - 1$, the vertex u' is adjacent to a vertex w of G which does not belong to F .
7. Therefore, T is isomorphic to the subgraph of G obtained by adding the vertex w and the edge $u'w$ to F .

Wiener index

Definition (Wiener index)

Average distance, instead of maximum distance, between a pair of vertices can be obtained by summing the distances between every pair of vertices in a graph divided by the number of vertex pairs, $\frac{n}{2}$. This is equivalent to finding $D(G) = \frac{1}{2} \sum_{u,v \in V(G)} d(u, v)$ for graph G . The value $D(G)$ is called *Wiener index*.



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Among trees with n vertices, the Wiener index, $D(T) = \frac{1}{2} \sum_{u,v \in V(T)} d(u, v)$, is minimized by stars and maximized by paths, both uniquely.

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Theorem

Among trees with n vertices, the Wiener index, $D(T) = \frac{1}{2} \sum_{u,v \in V(T)} d(u, v)$, is minimized by stars and maximized by paths, both uniquely.

Sketchy Proof

1. A tree, T , of order n has $n - 1$ edges. So, it has $n - 1$ pairs of vertices at distance 1, and all other pairs have distance at least 2.
2. Consider a leaf x in T , and let v be its neighbor.

Some more

Sketchy Proof

3. If all other vertices have distance 2 from x , then they must be neighbors of v , and thus T is the star.
4. The value of Wiener index is,

$$D(T) = (n-1) + 2 \binom{n-1}{2} = (n-1)^2$$

5. For the maximization, consider first $D(P_n)$. This equals to the sum of the distances from an endpoint u to the other vertices, added with $D(P_{n-1})$.
Now,

$$\sum_{v \in V(P_n)} d(u, v) = \sum_{i=0}^{n-1} i = \frac{n(n-1)}{2}$$

Some more

Sketchy Proof

6. Therefore,

$$D(P_n) = D(P_{n-1}) + \frac{n}{2}$$

7. Using Pascal's formula,

$$D(P_n) = \frac{n+1}{3}$$

8. Now, using induction on n , show that among all n -vertex tree, P_n is the only tree that maximizes $D(T)$.
9. **Basis:** $n = 1$. The only tree with one vertex is P_1 .
10. **Induction:** $n > 1$. Let u be a leaf of the tree T .

Some more

Sketchy Proof

11. So,

$$D(T) = D(T - u) + \sum_{v \in V(T)} d(u, v)$$

12. By hypothesis, $D(T - u) \leq D(P_{n-1})$, with equality if and only if $T - u$ is a path.

13. Thus it suffices to show that $\sum_{v \in V(T)} d(u, v)$ is maximized only when T is a path and u is an endpoint of T .

14. Now, list of distances from u . In P_n , this list is $1, 2, \dots, n-1$, all distinct, provided P_n is a path. So, the set of distances from u to other vertices has no gaps.

15. Otherwise, any repetition makes $\sum_{v \in V(T)} d(u, v)$ smaller.